Signals recap ~~of~~ Lecture 2

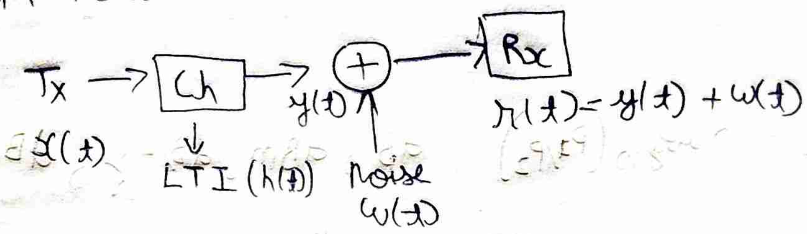
- Why do we need complex valued signal? When signals are real? → are transported

~~because analysis~~ ~~then~~ because analysis ~~of~~ becomes easier.

$x(t) \longrightarrow X(\omega)$ → has a complex representation
↳ complex

- Sampling of continuous signals is lossless if signal is bandlimited.

- After sampling, you do quantization (Acts as a channel) which introduces random noise
↳ additive noise

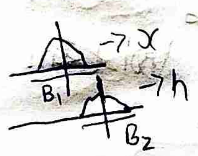


- noise is gaussian due to central limit theorem.

- We assume channel is LTI, so we can charecterize it ~~by~~ using $h(t)$.
Why is it this assumption valid? (It is an approximation)

- If $y(t) = \underset{B_1}{x(t)} * \underset{B_2}{h(t)}$ → what is the bandwidth of $y(t)$

$y(t) \le \min(B_1, B_2)$



- We assume we have Energy ~~…~~ signals and not Power signals in this course

→ Page 61 of textbook

- ~~…~~ f = frequency, ~~…~~ $|X(f)|^2$ → energy spectral density
↳ used to reduce noise in or something on oscilloscope
↳ check again

in a band $\int_{f_1}^{f_2} |X(f)|^2$

- TDMA (time division Multiplexing), TDD
↑ access
- FDMA (freq division Multiplexing)
- SDMA (Space) → antenna directivity → ~~cell~~ cellular systems

- we want to ~~…~~ reduce energy spectral density in other bandwidths
↳ for example in FDM ~~it~~ it might ~~interfear~~ interfear with other bandwidth channels where other people may be communicating

- In all of these Multiplexing we want to ensure the signals transmitted are orthogonal.

- If some thing is orthogonal in time is it also in freq → check
↳ $\int x_1(t)\, x_2^*(t)\, dt = 0$

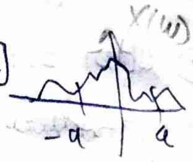
- CDMA → Code division Multiplexing → phase difference (sin, cos)

Band limited Signals → Uncertainity Principle

- we can never get Band limited signals → means ∞ time domain $x(t)$
- we can only generate time limited signal → ∞ ~~spectro~~ spectrum component $X(\omega)$

Definitions of bandwidths ↓

- 90% band width → 90% of energy spectral density $[-a, a]$



- Decible : $10 \log_{10}(P_1/P_2)$, dB , dBm , dB/Hz , dBC
  - dBm ↓ mW
  - dBC ↳ carrier freq

- 3dB bandwidth → where our power becomes half

- ~~Rayl~~ Paley-Wiener Criterion , → to check if a filter is realizable

$$\int_{-\infty}^{\infty} \frac{\ln|H(\omega)|}{1+\omega^2} < \infty$$

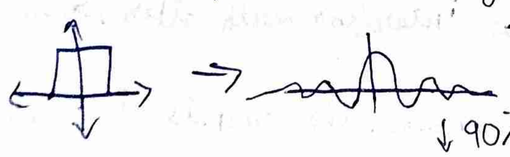
~~New~~ Lecture 3

- Time varying Convolution → for LTV systems.
- We model communication systems as

$$\underset{\text{reciever}}{r(t)} = \underset{\text{channel}}{h(t)} * \overset{\text{message}}{m(t)} + \underset{\text{AWGN}}{n(t)}$$

- We use orthogonality to seperate carrier signals, as by taking their projection $\int (m_1(t) + m_2(t))\, m_2^*(t)\, dt$

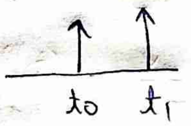
- In real life we can only generate Linear phase ~~systems~~ filters



- All types of filter that we ~~g~~ use we put them into the channel at the reciever or transmitter

- A simple wired channel can be ~~ll~~ modeled as
  $h(t) = a_0 \, \delta(t - t_0)$ → attenuation and delay due to distance

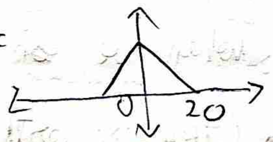
- but sometime ~~A~~ due to reflections from objects in a wired communication we get $h(t) = a_0 \delta(t - t_0) + a_1 \delta(t - t_1) \cdots$

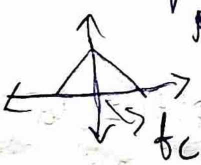
  we get let say  → in time we get maxima and minima

- This is called Inter Symbol Interfearence (ISI)
- We can construct Ideal filters in digital domain → Why ? because we made an assumption that $x[n]$ is periodic, (but we look at one period) so we see ~~[illegible]~~ sharp spectrum.
  → Check

Lecture 4

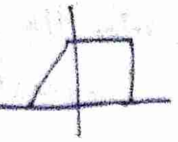
Recap
→ we ~~g~~ use GHz freq because of antenna length constraints.

- $M(\omega) =$  → we want to convert this to a higher freq for transmission → Up conversion

-  $= M(f - f_c)$ → carrier frequency

- The equivalent representation is $m(t) e^{j 2\pi f_c t}$

- $m(t)$ can be complex as well, $m(t) = m_{re}(t) + j m_{img}(t)$
  advantage, for a real signal the spectrum is symmetric so the info on one side is redundant  redundant so you can combine signal, $m_1 =$  $m_2 =$ 
  $\therefore \; m_1 + j m_2 =$ 

• $M(f) =$ 

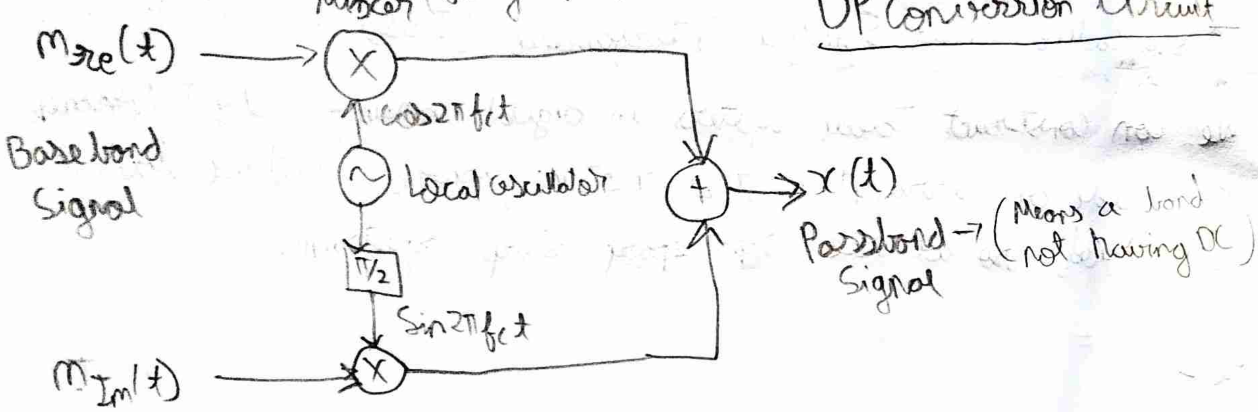
Scale $\frac{1}{2}M(-f\ f_c)$: $\frac{1}{2}M(f-f_c)$

for $\underset{\uparrow\ x(t)}{Re}\{m(t)e^{j2\pi f_c t}\} = \dfrac{z(t)+z^*(t)}{2}$, ~~s.t.~~ $z(t) = m(t)e^{j2\pi f_c t}$

$M^*(t) \longrightarrow m_{re}(t)\cos 2\pi f_c t + m_{Im}(t)\sin 2\pi f_c t$

$x(t) = m_{re}(t)\cos 2\pi f_c t - m_{Im}(t)\sin(2\pi f_c t)$

UP Conversion Circuit



• $\cos 2\pi f_c t \perp \sin(2\pi f_c t)$ (orthogonal) , when we say modulating we are modulating amplitude of both of them (they are orthogonal) ask again

• The phase is also being modulated

• We define an envelope $e(t) = \sqrt{m_{re}^2(t) + m_{Im}^2(t)}$, $\phi(t) = \tan^{-1}\left(\dfrac{m_{Im}(t)}{m_{Re}(t)}\right)$

$\therefore m_{re} = e(t)\cos\phi(t),\ m_{Im} = e(t)\sin\phi(t)$

Substitute in $x(t)$

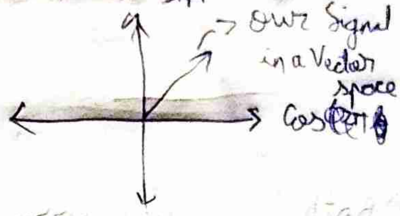
to get $x(t) = e(t)\cos(2\pi f_c t + \phi(t))$

- m_{Re} and m_{Im} can be arbitrary in time domain

- In freq's $f_c >> B$ because high bandwidth will spread to a freq which ~~as~~ our antenna will not transmit, so we have this restriction.

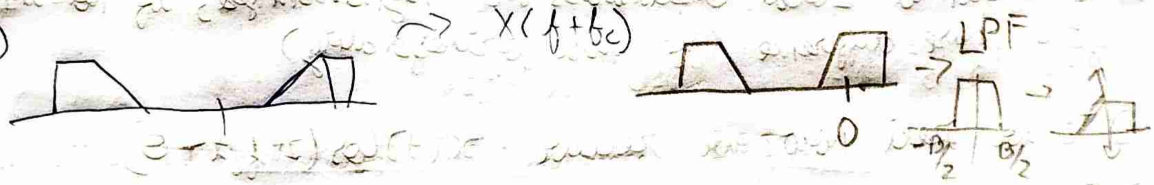
- $m(t) = m_I(t) + j\, m_Q(t)$

m_I → Inphase, m_Q → Quadrature Phase



- Now we have to ~~invert~~ invert this signal

$X(f) \rightarrow X(f+f_c)$



- But IRL we cannot multiply with imaginary freq

we do $Re\left(\underbrace{2}_{\text{Scaling}} x(t) e^{j2\pi f_c t}\right)$

$\Rightarrow 2x(t)\cos 2\pi f_c t = (1+\cos 4\pi f_c t)\, m_I(t) - m_Q(t) \sin 2\pi f_c t$ $\sin 4\pi f_c t$?

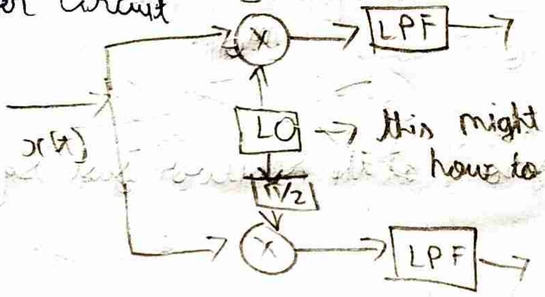
LPF $\rightarrow m_I(t)$

to get other component, ~~$2x(t)\cos($~~ ~~$-2\cos$~~

$-2x(t)\sin 2\pi f_c t =$ ~~$-2m_I(t)\sin(2\pi f_c t)\cos(2\pi f_c t)$~~

Receiver Circuit

Down Conversion Circuit



→ this might be out of phase with the transmitter how to deal with it?

↳ Because we can have finite duration signals, our spectrum will be ∞, so we ensure 99% of energy in the region of interest. But if we have multiple antennas talking then 1% from each may add up.

Lecture 5

- If we use wired medium we are sending only 1 signal, are we sending less information ~~comp~~ compared to wireless.?
  No as in wireless we use more bandwidth to send 2 signals

- Both the upconversion and downconversion blocks are exactly the same but how are we able to recover the signal. → ?
  ↳ Think about this

- We assumed Local oscillators are synchronized, if not then we get a phase difference (we loose orthogonality)

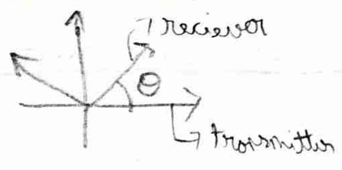
- ~~If we~~ ~~reci~~ now we recieve $x(t)\cos(2\pi f_c t+\theta)$
- transmit power amplifier :- multiply $x(t)$ at the transmitter by 2 to avoid scaling at the sender (noise is amplified)

$$\to m_I(t)\left[\cos(4\pi f_c t+\theta)+\cos\theta\right]$$
$$-m_Q(t)\left[\sin(4\pi f_c t+\theta)+\sin\theta\right]$$

after ~~low~~ LPF, $I \to m_I(t)\cos\theta - m_Q(t)\sin\theta$

$Q \to m_I(t)\sin\theta + m_Q(t)\cos\theta$

we recieve a rotated axis at the reciever



$$\begin{bmatrix}\cos\theta & -\sin\theta \\ \sin\theta & \cos\theta\end{bmatrix}\begin{bmatrix}m_I \\ m_Q\end{bmatrix}$$

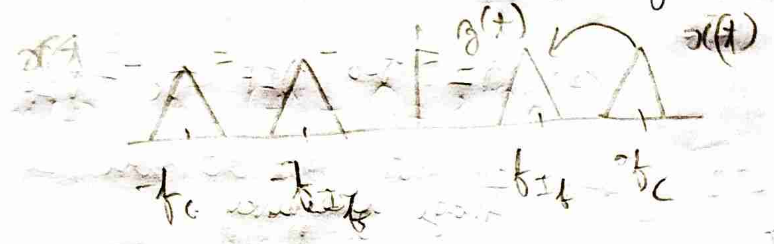
- Amplitude modulation, set M_Q to zero, at the reciever just square and add to get $m_I(t)$

- Phase modulation, set $e(t)$ to 1 and only change $\phi(t)$, derivative of $\phi(t)$ is the freq (freq modulation)

↳ we can use ~~phased~~ phase locked loops to synchronize

Hetrodyne - type or Super hetrodyne

- ~~...~~ Our phones have too many bands of ~~...~~ freq ~~...~~ that they use. We get the carrier freq to an intermediate freq which is what our circuits are designed for.



$$z(t) = Re\{x(t)\, e^{j2\pi(f_c - f_{If})t}\}$$

do

- There was an issue with circuit that ~~freq~~ at $f_c - 2f_{IF}$ was also shifted to the ~~...~~ f_{If} freq, ~~...~~ put a ~~BPF~~
In old AM radio ~~stations~~ they could hear 2 stations simultainiously.
- ~~At the~~ end you put a BPF at f_{If}

Lecture 6

- If LO are synchronized then you calls it homodyne
- Superhetrodyne → Bringing down to intermediate frequency and then bring that down again to ~~...~~ DC.
- ~~...~~ is around 450 MHz
I_{fc}
- at the reciever side we have $x(t)\cos(2\pi f_{LO} t)$

$$m_I(t) = \frac{m(t)}{2}\left[\cos(f_{LO}+f_c)t + \cos 2\pi(f_c - f_{LO})t\right]$$

$$m_Q(t) = \frac{m(t)}{2}\left[\sin 2\pi(f_c + f_{LO})t + \sin(2\pi(f_c - f_{LO})t)\right]$$

we get $f_c - f_{LO} = f_{IF}$ → we convert to ~~...~~ f_{If}

• We saw that this freq of f_{LO} we had issues

this interfears

↓

this is for $Re\{x(t)e^{-j2\pi f_{LO}}\}$

$f_2 - f_{LO} = -f_{IF}$

$f_2 = f_{LO} - f_{IF} = f_c - 2f_{IF}$

↓

Image frequency.

• here we are shifting to the ~~right~~ left, but we can also shift to the right.

↳ $f_c + f_{LO} = f_{IF} \Rightarrow f_{LO} = f_c + f_{IF}$

we want  we get 

• Why is it getting flipped? → see complex representation

$\boxed{\text{Spectral inversion}}$

$f_{LO} = f_c - f_{IF}$

$\frac{m_I(t)}{2}\cos$

$- m_Q(t)\sin$

$= \frac{1}{2} Re\{m(t)e^{j2\pi f_c t}\}$

$f_c + f_{IF}$

$\frac{m_I(t)}{2}\cos$

$+\frac{m_Q(t)}{2}\sin$ → Check

$= \frac{1}{2} Re\{m^*(t)e^{j2\pi f_c t}\}$

• How to select from the above two

we had image frequency as $f_c - 2f_{IF}$ before and now shifting right we have $f_c + 2f_{IF}$

↳ so we can use a low pass filter

it is an Image ~~Reflection~~ Rejection Filter.

• lets say $f_c \in [f_l, f_u]$ → same antenna, different f_c operation

picking LO, 1

$f_{LO} = [f_l - f_{IF}, f_u - f_{IF}]$

$= [f_l + f_{IF}, f_u + f_{IF}]$

→ $\frac{f_l - f_{IF}}{f_u - f_{IF}}$

↓

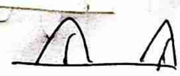
the ratio will be different

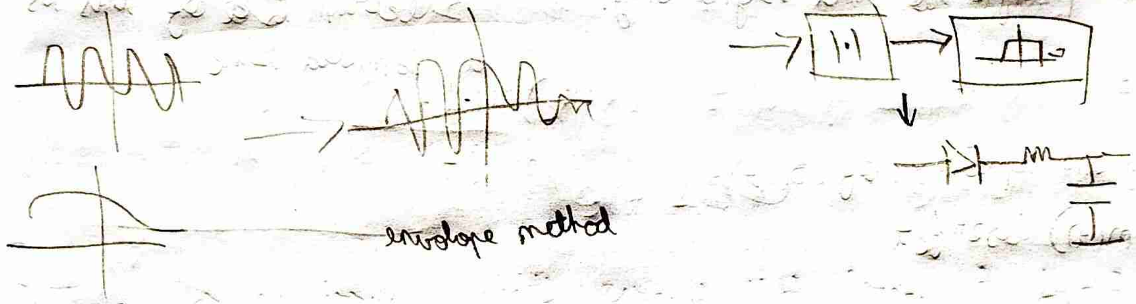
- That ratio is the swing of the filter and we want it to be small, small, even if the range of freq is ~~diff~~ the same.

Amplitude Modulation

- $x(t) = e(t)\cos(2\pi f_c t + \phi(t))$
- Let's say we don't want to worry about θ (Phase offset between filter). We can just set $m_Q(t) = 0$

then $x(t) = m_I(t)\cos(2\pi f_c t)$

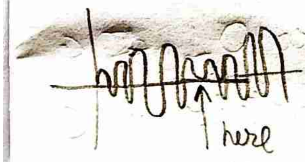
- AM is very inefficient, what we transmit  we are sending 4 times of what we need



- Issues with AM $\Rightarrow$ $m_I(t)$ cannot be -ve as |.| ignores it.

Lecture 7

Phase reversal issue, for -ve $m(t)$ there is a ~~phase~~ signal gets a discontinuity



DSB-SC

Double side band ~~or~~ Suppressed Carrier

- we ~~can~~ can add an offset to $m(t)$

↳ $(m(t) + A_c)\cos 2\pi f_c$, with $m(t) + A_c \geq 0$ always

↓ Amplitude of the carrier freq (DSB-AM) ↗ Amplitude Modulation

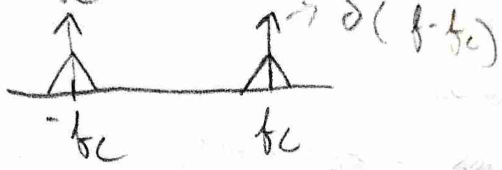
- If $\min m(t) = -M$, then modulation index $\alpha = \frac{M}{A_c} \leq 1$

↳ Also assume $M(t)$ has no DC components

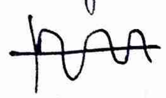
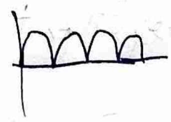
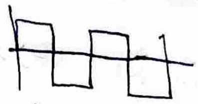
$\therefore\ A_c(1 + \alpha\, m(t))\cos(2\pi f_c t)$

- DSB-SC spectrum

- DSB-AM spectrum

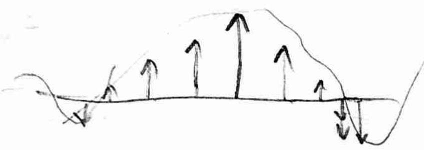


- the modulus is not an LTI system, but how do you see it mathematically

$x(t) \rightarrow$  , $|x(t)| =$  equivalent to multiplying with 

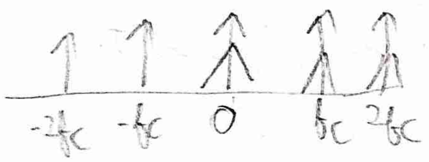
which is specific to this signal only.

Spectrum of this is a sampled sine



so what is the spectrum of $x(t)$

$A_c(1+\alpha m(t)) \cos 2\pi f_c t$



- How insufficient is our AM? How to quantify it?

↳ $\dfrac{\text{Power of message}}{\text{Power of Tx signal}}$

$$x^2(t) = A_c^2(1+\alpha m(t))^2 \cos^2(2\pi f_c t) \rightarrow \text{message} + \text{Carrier}$$

message: $A_c \alpha m(t)$, Carrier: A_c

$$= \underbrace{\frac{A_c^2}{2}(1+\alpha m(t))^2}_{y(t)} (1+\cos(4\pi f_c t))$$

$y(t) \cos 4\pi f_c t$ after integrating is zero.

- Also our assumption that the bandwidth of message is less than f_c so we will definitely not get a DC component $\rightarrow$ Integral is zero

◦) the power of $x(t)$ is then $\frac{A_c^2}{2}(1+\alpha^2 \underbrace{P_m}_{\text{Power of } m(t)})$ zero

◦) Is ~~energy~~ Power in Passband the same as in the power in the baseband?

↳ ~~part is~~ but it doesn't make AM or SC any efficient

◦) $x_m^2(t) = (A_c \alpha m(t) \cos 2\pi f_c t)^2$

this is in the Passband $= \frac{A_c^2}{2}\alpha^2 m^2(t)$

$$\eta = \frac{\text{Power of msg}}{\text{Power of } T_x \text{ signal}} = \frac{\frac{A_c^2}{2}\alpha^2 P_m}{\frac{A_c^2}{2}(1+\alpha^2 P_m)} \approx \frac{a}{1+a}$$

has to be normalized

→ can never be 100%

η → Power efficiency

PAPR = ~~Peat~~ $\frac{\text{Peak of } m(t)}{\text{Avg}^{\text{r}} \text{ of } m(t) \text{ Power}}$ → it is energy and not Power (Misnomer)

PAPR → Peak to average Power Ratio

◦) maximum efficiency is 50%, but for this we assume $P_m = 1$ and $\alpha = 1$, but if you send a sinusoid, $P_m = 1/2$ and then you get even lower efficiency. this just ~~mea~~ shows there is a relation to PAPR

Lecture 8

◦) we had assumed, $\int m(t) = 0$, no DC component.

◦) How does the PAPR affect the whole of above setup.

$$\text{PARP} = \frac{\frac{A_c^2}{2}(1+1)}{\frac{A_c^2}{2}(1+\alpha^2 P_m)} \rightarrow \frac{2}{1+\alpha^2 P_m} = b$$

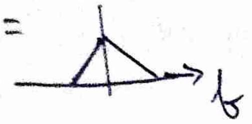
High PAPR is always bad

Power in $A_c(\alpha m(t)+1)\cos 2\pi f_c t$

now sub this in $\eta = 1 - \frac{b}{2}$, ↑PAPR, ↓η

SSB (Single side band)

removing redundancy.

$M(f) =$ 

$$M(f) = M(f)u(f) + M(f)u(-f)$$
$$M_+(f) + M_-(f)$$
$$\downarrow \qquad \downarrow$$
$$m(t) = m_+(t) + m_-(t)$$ because fourier transform is linear

multiplying by $u(f)$ is convolution in time
↳ find out

$$X(f) = M(f-f_c)u(f-f_c) + M(f+f_c)u(f+f_c)$$

$$\therefore x(t) = m_+(t)e^{j2\pi f_c t} + m_-(t)e^{-j2\pi f_c t}$$

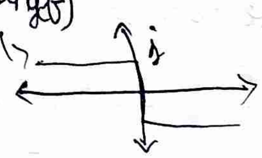
$x(t)$ is real signal → verify, $X(f) = X^*(-f)$, ~~then~~

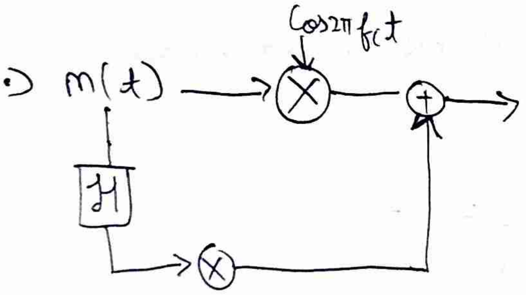
$$x(t) = m_+(t)\left[\cos 2\pi f_c t + j\sin 2\pi f_c t\right]$$
$$m_-(t)\left[\cos 2\pi f_c t - j\sin 2\pi f_c t\right]$$

$$= \underbrace{m(t)\cos 2\pi f_c t}_{\text{In phase component}} - \underbrace{\overbrace{(-j m_+(t) + j m_-(t))}^{m_h(t)}\sin 2\pi f_c t}_{\text{Quadrature}}$$

$\hookrightarrow m_h(t) = \mathcal{H}(m(t))$ → ~~Impulse~~ frequency response $-j\,\mathrm{sign}(f)$

In frequency $-jM_+(f) + jM_-(f)$





find inverse transform of $-j\,\mathrm{sign}(f) \longleftrightarrow \frac{1}{\pi t}$

•) ~~We can see to send our message comp!~~

•) Now analyse $(m(t)+A_c)\cos 2\pi f_c t - m_h(t)\sin 2\pi f_c t$

$$= e(t)\left[\cos(2\pi f_c t + \phi(t))\right]$$

$$e(t) = \sqrt{(m(t)+A_c)^2 + m_h^2(t)} \approx A_c + m(t)$$ → hilbert component being negligible makes us use envolope method seen previously ~~to demodulate~~ the signal ↓ In SSB

→ check

•) $m(t)$ is bandlimited so $m_h^2(t)$ is not unbounded

•) ~~The demodulation circuit~~ is the same as one we saw for Down conversion, we get $m(t)$ in the inphase arm, we are just sending a different looking signal.

QAM → Quadrature Amplitude modulation

↳ 2 AM signals which are orthogonal, this is widely used

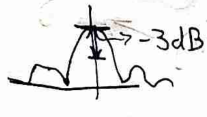
↳ Amplitude shift keying → Write from Screen

Assignment 1

•) ↳ find out why the spectrum was comming up with imaginary ?

↳ indexing of fft was wrong. you had to shift the time domain signal too

•) 3dB bandwidth → log plot of energy Spectral density, look at 3dB below the peak amplitude

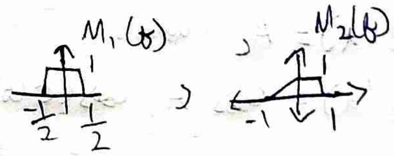


•)

Assignment 2

(I) Upconversion

Generate baseband signal by taking iFFt of



Upconvert to $f_c = 10\,Hz$

plot the Converted time and frequency

2) Down conversion

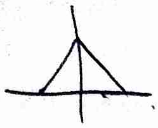
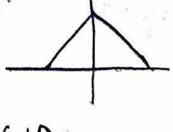
Downconvert your upconverted signal.

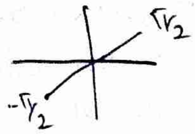
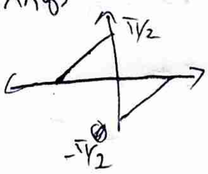
- QAM $\rightarrow x(t) = A_c[m_I(t)\cos 2\pi f_c t - m_Q(t)\sin 2\pi f_c t]$

Hilbert transform

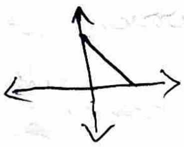
$H(x(t)) = x(t) * \frac{1}{\pi t}$

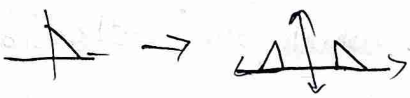
$X_h(f) = X(f)(-j\,\mathrm{Sign}(f))$

$|X(f)|$  $|X_h(f)| =$ 

$\measuredangle X(f)$  $\rightarrow$ $\measuredangle X_h(f)$ 

- Spectrum of $x(t) + j x_h(t)$ $\rightarrow$ analytic representation of a signal

$= X(f) + X(f)\,\mathrm{Sign}(f)$ $\rightarrow$ 

- this is a technique to remove -ve freq values in spectrum
- After this you upconvert it, 

- Section 2.8.3 about complex baseband is interesting
  - in text book

Angle Modulation

- We will work on $\phi(t)$, and we have 2 options

Phase Modulation (PM) $\boxed{\phi(t) = k_p m(t)}$ $\rightarrow$ deviation constant

freq Modulation (FM) $\rightarrow$ $\boxed{f(t) = \frac{1}{2\pi}\frac{d\phi}{dt} = k_f m(t)}$

$\hookrightarrow$ $\boxed{\phi(t) = 2\pi k_f \int_0^t m(\tau)\, d\tau}$

if $e(t) = A_c$ (constant)

$m_I(t) = A_c \cos(\phi(t))$

$m_Q(t) = A_c \sin(\phi(t))$

- ~~After~~ If LO is not synchronize we get

$\cos\theta \cos\phi(t) + \sin\theta \sin\phi(t)$ in the I arm we receive $\cos(\phi(t) - \theta)$ and $\sin(\phi(t) - \theta)$ ← Phase delay

we can do $\cos^{-1}$ it to recover phase ⇒ which contains our message

- this is good for PM, but ~~I~~ how do you recover for F.M

Assumption $m(t)$ has no DC component

- you derivate $\frac{1}{2\pi} \frac{\partial}{\partial t}(\phi(t) - \theta) \Rightarrow \frac{1}{2\pi} \frac{d}{dt}(\phi(t)) \rightarrow$ so θ just goes away.

so we prefer FM

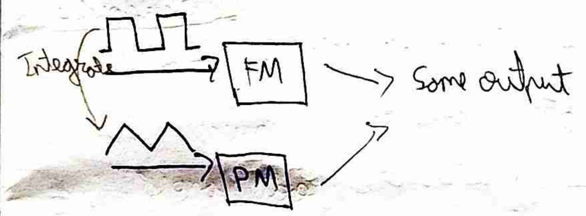
- Circuits for Both FM and PM ~~dodu~~ modulations work for each other

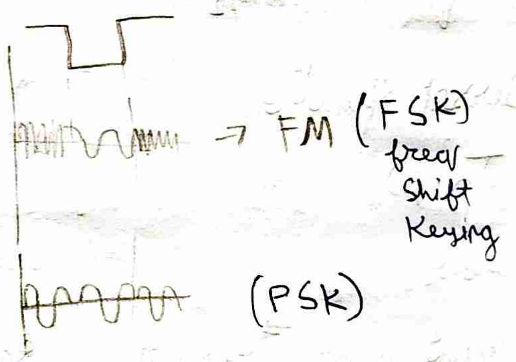
$m(t) \rightarrow \boxed{FM} \rightarrow X_{FM}(t)$

$m(t) \rightarrow \boxed{\frac{d}{dt}} \rightarrow \boxed{FM} \rightarrow X_{PM}(t)$ → $K_p = 2\pi K_f$

$m(t) \rightarrow \boxed{\int dt} \rightarrow \boxed{PM} \rightarrow X_{FM}$

- What happens to the spectrum after modulation?





- Satellite ~~using~~ use FSK, digital communication ~~uses~~ uses PSK

- Modulation index $\beta = \frac{k_f \max(m(t))}{B} = \frac{\Delta f}{B}$

multiply

→ max freq division

B → Bandwidth of the message

B → Bandwidth

For FM

$= k_p \max(m(t))$ For PM

Max deviation of freq from the carrier freq

• looking at spectrum of $\boxed{FM}$, PM is the Same

• Narrow band FM : → Δf is very small , $\phi(t) << 1$

$$A_c\left[\underbrace{\cos\phi(t)}_{\approx 1}\cos 2\pi f_c t - \underbrace{\sin\phi(t)}_{\approx \phi(t)}\sin 2\pi f_c t\right]$$

this is again similar to A.M

$X_{FM}(f)$ has bandwidth as $2B$, we assume bandwidth is same even after integration

• We have to use spectrum of $Re\left\{e^{j2\pi f_c t + j2\pi k_f \int_0^t m(\tau)\,d\tau}\right\}$

Say $m(t) = \cancel{\cos(2\pi f_m t)}\ \cos(2\pi f_m t)$ → Band width $B = f_m$

we have $\beta = \dfrac{k_f \times 1}{B}$ (modulation index) →

$e^{j\beta \sin 2\pi f_m t}$

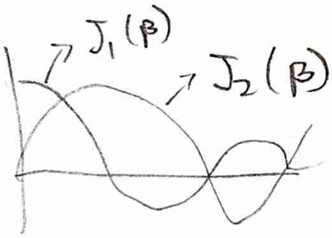
• This signal is periodic

$$\therefore C_n = \frac{1}{2\pi}\int_{-\pi/2}^{\pi/2} e^{j(\beta\sin\omega - n\omega)t}\,dt \qquad 2\pi f_m = \omega$$

Bessel function

$J_n(\beta)$ →

order



• What we are transmitting $Re\left\{e^{j2\pi f_c t}\sum J_n(\beta)e^{j2\pi f_m t}\right\}$

$$\boxed{= \sum J_n(\beta)\cos(2\pi f_c t + n2\pi f_m t)}$$

Assignment - 2 , Q7

• Message Value goes to ∞

• do we want $\beta << 1$

• Do we want lesser bandwidth for FM because compression

•) New class - the fourier series of ~~e~~ rect

$c_n = \frac{1}{1/f_m} \int e^{j\beta \sin 2\pi f_m t} e^{-j 2\pi n f_m t} dt \qquad V = 2\pi f_m$

$\Rightarrow J_n(\beta) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{j(\beta \sin V - nV)} dV$ → n^{th} order, 1st kind bessel function

•) This doesn't depend on f_m? wrong these are just the coefficient of fourier series, finally we have $\sum_{n=-\infty}^{\infty} J_n(\beta) e^{j2\pi f_m n}$

•) $J_n(\beta) \approx \frac{(\beta/2)^n}{n!}$ for $\beta << 1$

•) This decays very fast, so let us try to get the 99% bandwidth of the signal from the coefficients

$99\% \text{ BW} = \sum_{n=-K}^{K} J_n^2(\beta)$

$= J_0^2(\beta) + 2\sum_{n=1}^{K} J_n^2(\beta)$ → there is a symmetry

•) Bessel functions appear in a lot of places → drum surface vibrations, and a lot of places → find

•) ~~Bandwidth ≈ 2(K+1)~~

•) Carson's Rule | $K \approx \beta + 1$ →

•) ∴ Bandwidth ≈ ~~2(K+1)~~ $2(\beta+1) f_m$ → Bandwidth of message

•) So there will be some distortions, we do not ~~sent~~ send $e^{j\beta \sin f_m t}$ but $\sum_{n=-K}^{K} e^{j2\pi n f_m}$

•) How do you modulate FM signals → Voltage Controlled Oscillator (VCO)

•) say $f(t) = K_v V(t) + f_c$, we get $\cos(2\pi f_c t + 2\pi K_v \int_0^t V(\tau) d\tau)$

derivate it

•) How do you ~~de~~ demodulate the signal, you differentiate it to get the freq outside

we get $A_c \sin(\dots) \times (2\pi k_f m(t) + 2\pi f_c)$

Δf the frequency deviation should not be large to ~~ded~~ demodulate

↳ this is now similar to AM, and ~~so~~ we can use that ~~to~~ system

•) For P-M the analysis is still the same, we get a shift in $J_n(\beta)$

•) But ~~the~~ bandwidth changes

$BW_{FM} \approx 2(\Delta f + f_m)$ | $BW_{PM} \approx 2(\Delta\phi + 1) f_m$ → Check

•) For the same message FM has a ~~lot~~ lesser bandwidth

•) FM or PM is spectrally inefficiency compared to ~~A~~ AM, but in the power efficiency it is almost 100% in FM, this becomes significant when noise ~~is~~ is introduced (we want high SNR)

New class

$\beta_{PM} = k_p \max|m(t)|$, $\beta_{FM} = \dfrac{k_f \max|m(t)|}{B}$

•) We want a better demodulator than a differentiator for FM

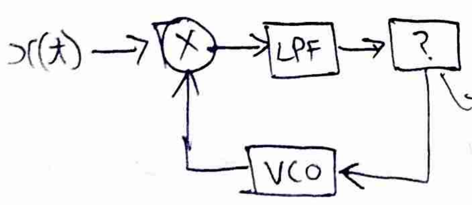
Phase Locked Loops PLL

$x(t) = A_c \cos(2\pi f_c t + \phi(t)) \longrightarrow (\times) \longrightarrow \dfrac{A_c A_v}{2}\left[\sin(4\pi f_c t + \phi(t) + \phi_v(t)) + \sin(\phi(t) - \phi_v(t))\right]$

removed from LPF

VCO: $-A_v \sin(2\pi f_c t + \phi_v(t))$

→ see last page



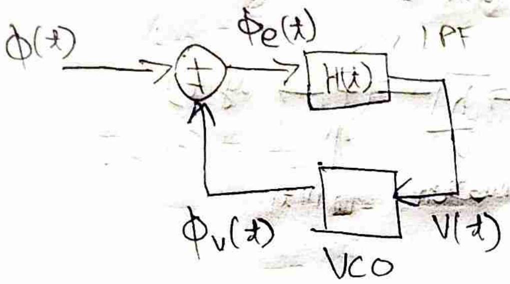
↳ we drive the error to zero

↳ How, we have $\dfrac{A_c A_v}{2}(\sin(\phi(t) - \phi_v(t))$

if this goes to zero then, $\phi_v(t) \approx \phi(t)$ our ~~ded~~ demodulated signal

our message is what is present in $\phi_v(t)$ → Next page first order PLL

- Assumption $\approx \frac{A_c A_v}{2}[\phi(t) - \phi_v(t)]$ if error is small



as $f(t) = f_c + k_v V(t)$ for the VCO

- $\phi_e(t) + \phi_v(t) = \phi(t) \Rightarrow \frac{d\phi_e}{dt} + K_v V(t) = \frac{d}{dt}\phi(t)$

$$\Rightarrow \frac{d\phi_e(t)}{dt} + 2\pi K_v h(t) \times \phi_e(t) = \frac{d\phi(t)}{dt}$$ Fourier transform

$$j2\pi f\, \phi_e(f) + 2\pi K_v H(f) \times \Phi_e(f) = 2\pi f j \Phi(f)$$

$$\Rightarrow \Phi_e(f) = \frac{\Phi(f)}{1 + \frac{k_v H(f)}{jf}}$$

- $V(f) = H(f)\Phi_e(f)$

$$= \frac{H(f)}{1 + \frac{k_v}{jf} H(f)} \Phi(f)$$

assume $|H(f)| >> 1$ (put an amplifier)
inside the bandwidth of the message $\Rightarrow$ So we can use a LPF insted of an amplifier, Gain of LPF should be high

$$\therefore V(f) = \frac{jf}{K_V}\phi(f)$$

$$V(t) = \frac{1}{2\pi K_v}\frac{d\phi(t)}{dt} = \frac{K_f}{K_v} m(t)$$ $\Rightarrow$ $V(t)$ is proportional to our message

↳ this is a PI controller, VCO is the integrator block

- Pg 128 of text book has a lot more content on PLL

New class

- PLL is unstable $\Rightarrow$ the message should be bandlimited
- If our message is has a DC component, the integrator causes the system to become unstable

•) Would you pick AM or FM?

•) Let us look at Additive Noise in channels

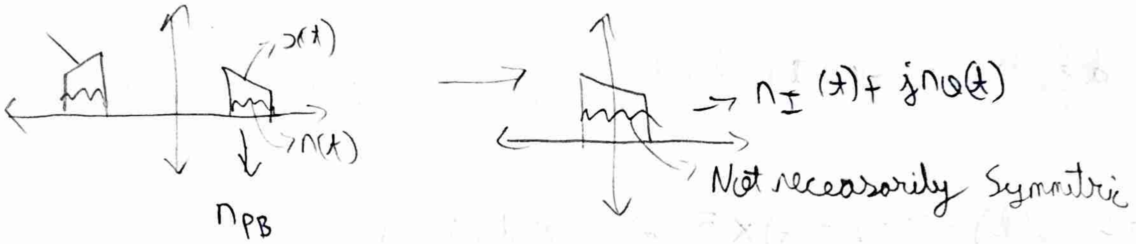
$y = hx + n$

$\boxed{T_x} \to (\text{channel}) \to \boxed{R_x}$

$x(t) \to \boxed{h} \to \oplus \to y(t)$

attenuation $\qquad n(t)$

•) There is noise in baseband and passband, but the passband noise is real, but baseband noise ~~~~ is not necessarily real



•) Our LPF determines the quality of noise effect.

•) In AM, if

			Noise	
	AM	FM	AM	FM
$M_{demod}\ I(t) =$	$m(t)$	$\cos\phi(t)$	$m(t) + n_I(t)$	$\cos\phi(t) + n_I$
$M_{demod}\ Q(t) =$	0	$\sin\phi(t)$	$n_Q(t)$	$\sin\phi(t) + n_Q$

•) demodulation is simpler for FM → take derivative

at the reciever lets say we have

$\cos\phi(t) + n_I(t) + j(\sin\phi(t) + n_Q(t))$

combine terms

we get $e^{j\phi(t)} + r(t)\, e^{j\theta(t)}$ → •) we project this signal into the Unit circle in the Complex Plane,

Our message $e^{j\phi(t)}$ will always be found there while noise can be anywhere.

•) We do this as our message signal is contained in the phase.

New Class

- FM has a better power efficiency → which makes it ~~[illegible]~~ robust to noise

- $\underset{R_x}{y(t)} = \underset{\text{attenuation}}{h}\ \underset{T_x}{x(t)} + \underset{\text{noise}}{n_{RF}(t)} \rightarrow n_I(t) + j\,n_Q(t)$

Random Variables

$X(t) \rightarrow$ random variable

↳ Sample Space, $F \rightarrow$ event Space, Probability Measure

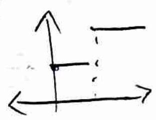
Ω $\{H, T\}$ 2^{Ω} P ↳ P.D.F

$X : \Omega \rightarrow \mathbb{R}^n$

- Borel Set → Our message is a random variable, but ~~[illegible]~~ Ω is uncountably ∞ so you use this to ~~pick~~ model probablity

- PDF for a few random variable may not even exist, but CDF exists
  ↳ HW

- CDF → monotonic, Right Continious



$$F_X(x) = P(X \leq x)$$

- $\mathbb{E}[g(X)] = \int g(x) f_X(x)\, dx$

$$Var(X) = \mathbb{E}\{(X - E[X])^2\}$$

$$Cov(X, Y) = \mathbb{E}\{(X - \mathbb{E}(X))(Y - E[Y])\}$$

~~Cross~~ Cross Covariance

Cor → see

If ~~[illegible]~~ the Random variable is Complex you will do conjugate

$$Var(X) = E[(X - E[X])^*(X - E[X])^*] \rightarrow \text{this is real}$$

$$Cov(X, Y) = E[(X - E(X))(Y - E(Y))^*] \rightarrow \text{this can be complex}$$

- Bernoulli $\longrightarrow X \in \{x_0, x_1\}$ ($x_1 \rightarrow p$, $x_0 \rightarrow 1-p$) $Be(p)$

- Uniform $\longrightarrow X \sim U[a, b]$, $f_X(x) = \begin{cases} \frac{1}{b-a} \\ 0 \end{cases}$

• Gaussian $\rightarrow X \sim \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

• $f_{XY}(x,y) = f_X(x) f_Y(y) \rightarrow$ means Independent

$E[X.Y] = E[X]E[Y] \rightarrow$ Uncorrelated

• Gaussian R.V are ~~Uncorr~~ Uncorelated and Independence

$Z = \begin{bmatrix} X \\ Y \end{bmatrix}$, $\frac{1}{2\pi \det(\Sigma)} e^{-\frac{1}{2}(Z-\mu)^H \Sigma^{-1} (Z-\mu)}$

If $X_{n\times 1}$ $\rightarrow$ $2\pi^n$

$\Sigma = \begin{bmatrix} C_{ij} = Cov(X_i, Y_j) \end{bmatrix}$

$\begin{bmatrix} n_I(t) \\ n_Q(t) \end{bmatrix} = N$, $n_I(t) \rightarrow N(0, \sigma_I^2)$, $n_Q(t) \rightarrow N(0, \sigma_Q^2)$

$\longrightarrow \frac{1}{2\pi \underbrace{(\det \Sigma)^{1/2}}_{\sigma_I \sigma_Q}} e^{-\frac{1}{2}\underbrace{\left(\begin{bmatrix} n_I \\ n_Q \end{bmatrix}^T \Sigma^{-1} \begin{bmatrix} n_I \\ n_Q \end{bmatrix}\right)}_{\frac{n_I^2}{\sigma_I^2} + \frac{n_Q^2}{\sigma_Q^2}}}$

They are independent $\Sigma = \begin{bmatrix} \sigma_I^2 & 0 \\ 0 & \sigma_Q^2 \end{bmatrix}$

• If You transform N to polar Coordinates what happens to the distribution of $\begin{bmatrix} r \\ \theta \end{bmatrix}$ from $\begin{bmatrix} n_I \\ n_Q \end{bmatrix}$

~~$R = \sqrt{n_I^2 + n_Q^2}$~~ $R = \sqrt{N_I^2 + N_Q^2}$, $\Theta = \tan^{-1}\left(\frac{N_Q}{N_I}\right)$

• $Y = g(X)$, $f_Y(y) = \frac{f_X(g^{-1}(y))}{|g'()|} = \frac{f_X(g^{-1}(y))}{\det(J)}$ $\rightarrow$ for multivariable

$N_I = R\cos\Theta$

$N_Q = R\sin\Theta$, find Jacobian of this

- HW, find $f_{R,\Theta}(r,\theta) = ?$

New class

- Random Vector

Σ → Co-varience matrix

If $X \in \mathbb{R}^n$, $X \sim N(\mu, \Sigma)$

$$= \frac{1}{(2\pi)^{n/2}\det(\Sigma)} e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)}$$

If $X \in \mathbb{C}^n$ $$= \frac{1}{(2\pi)^n \det(\Sigma)} e^{-1/2 (x-\mu)^H \Sigma^{-1}(x-\mu)}$$

- If Σ is diagonal matrix → ~~xxx~~ distribution is independent so multiply the marginals to get joint P.D.F

- Yesterday we were doing $X \to Y$

$$\begin{bmatrix} n_I \\ n_Q \end{bmatrix} \quad \begin{bmatrix} r \\ \theta \end{bmatrix}$$

J → Pg 218 in textbook

- $$J = \begin{bmatrix} \frac{\partial r}{\partial n_I} & \frac{\partial r}{\partial n_Q} \\ \frac{\partial \theta}{\partial n_I} & \frac{\partial \theta}{\partial n_Q} \end{bmatrix} = \begin{bmatrix} \frac{n_I}{\sqrt{n_I^2+n_Q^2}} & \frac{n_Q}{\sqrt{n_I^2+n_Q^2}} \\ \cancel{\frac{-n_Q}{\sqrt{n_I^2+n_Q^2}}} \; \frac{-n_Q}{n_I^2+n_Q^2} & \frac{n_I}{n_I^2+n_Q^2} \end{bmatrix}$$

- you have to divide by ~~xxx~~ det(J) for your distribution

↳ If you are doing a transformation

$$\det(J) = \frac{1}{r}$$

- $$f_{r,\theta}(r,\theta) = \frac{r}{2\pi\sigma^2} e^{-1/2\left(\frac{r^2}{\sigma^2}\right)}$$

- $$f_r(r) = \int_{-\pi}^{\pi} (\ldots)\, d\theta = \frac{r}{\sigma^2} e^{-1/2\left(\frac{r}{\sigma}\right)^2} \longrightarrow \text{Rayleigh distribution}$$

- $$f_\theta = \frac{f_{r,\theta}}{f_r} = \frac{1}{2\pi} \rightarrow \text{they are uniform}, \; \Theta \sim \text{Unif}[-\pi,\pi]$$

$$R \sim \frac{r}{\sigma^2} e^{-\left(\frac{r}{\sigma}\right)^2}$$

$[1\ 0]$ → distribution in this direction

- $$[1\ 0] \begin{bmatrix} n_I \\ n_Q \end{bmatrix} \rightarrow \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{n_I^2}{2\sigma^2}}$$

- What about a distribution in any other direction → we get the same distribution → Symmetric around θ

- They are called, Circular Symmetric Complex Gaussian
- If $\sigma_Q \neq \sigma_I$ it is not symmetric
- $y(t) = h\,x(t) + n(t)$

our $n(t)$ is a random process $X_1, X_2 \cdots X_n \to$ a sequence
but we have continuous time random process, ∴ $X(t)$

- For any random variable we look at its statistics (Mean etc)
- For a random process? $\to X_{t_1} - X_{t_2}$

different time same statistic $\to$ strict sense Stationary process

In real life Wide sense stationary - (WSS)

1) mean is stationary $m_X(t) = \mathbb{E}\{X(t)\} = m_X$ (constant)

This is a signal deterministic

Auto correlation

2) $R_X(t_1, t_2) \to$ Correlation of random process $= \mathbb{E}[X_1(t_1)X^*(t_2)]$
$\hookrightarrow$ auto correlation

$= \int x_{t_1} x^*_{t_2} f(x_{t_1}, x_{t_2})\,dt_1\,dt_2$

depends on $t_1 - t_2$ not t_1 or t_2

if this $= R_X(t_1 - t_2) \to$ (WSS)
$\hookrightarrow$ Samples separated by same amount of time have a similar distribution

- $X(t)$ WSS $\to m(t)$, $\to R(\tau)$ we can see $R(-\tau) = R^*(\tau)$
$\hookrightarrow$ will have real spectrum

- Example of random process $X(t) = A_c \cos(2\pi f_c t + \Theta)$
Θ Unif $\sim [-\pi, \pi]$

$$m_X(t) = \mathbb{E}[X(t)] = \frac{1}{2\pi}\int_{-\pi}^{\pi} A_c \cos(2\pi f_c t + \theta)\,d\theta = 0 \to \text{it is stationary}$$

$\cdot)\ R_x(t_1, t_2) = \int \cancel{x(t_1)\ x^*(t_2)} \quad \mathbb{E}\{X(t_1) X^*(t_2)\}$

$$= \int A_c^{*} \cos(2\pi f_c t_1 + \theta)\, A_c \cos(2\pi f_c t_2 + \theta)\, dx_{t_1}\, dx_{t_2}$$

here θ is random | but it is fixed (doesn't change with time)

↳ this is WSS → How?

New class

↳ for analog systems we have continuous time random process

$$\underbrace{Y(t)}_{Rx} = h \underbrace{X(t)}_{Tx} + \underbrace{N(t)}_{AWGN}$$

$\cdot)$ WSS $\doteq$ $m_x(t) = m_x$ | $R_{xx} = R_{xx}(t_1 - t_2)$ ↗ lag or delay

$\cdot)$ Example from last class

$$X(t) = A_c \cos(2\pi f_c t + \theta)$$ ↳ Unif $[-\pi, \pi]$

$= g(\theta, t)$ → θ is fixed after being picked randomly but different t gives a new value → this is a random process

$\cdot)\ \mathbb{E}\{x(t)\} = ?$

↳ $\cdot\ \mathbb{E}\{g(\theta)\} = \int g(\theta)\, f_\theta(\tau)\, d\tau \qquad \phi = g(\theta)$

$= \int \phi\, f_\phi\, d\tau$ → Prove this

$$f_\phi = \frac{f_\theta(g^{-1}(\phi))}{g'} \quad f_\theta(\theta)?$$

from last page

$$E\{X(t)\} = \int_{-\pi}^{\pi} A\cos(2\pi f_c t + \theta)\frac{d\theta}{2\pi} = 0$$

$$R_{XX}\{X(t)\} = \iint A_c\cos(2\pi f_c t_1 + \theta)\, A_c\cos(2\pi f_c t_2 + \theta)\, f_X(x_{t_1}, x_{t_2})\, dx_{t_1}\, dx_{t_2}$$

- We are taking ensemble average ↑, the process is ergodic
  - → try transforming to $g(\theta)h(\theta)$ and integrate over θ
- We transform and integrate over θ

we get $R_{XX}(t) = \frac{A_c^2}{2}\cos(2\pi f_c(t_1 - t_2))$

- Say during transmission $X(t) = M(t)\cos 2\pi f_c t$

  $M(t)$ → Say it is WSS

$\mathbb{E}(X(t)) = \mathbb{E}\{M(t)\}\cos 2\pi f_c t$ → Periodic

- But at the receiver $X(t) = M(t)$ → which is again WSS
  - → We see blocks in Comsys change the stationary state

what is R_{XX} of this?

$$R_{XX}(t) = \mathbb{E}\{M(t)\cos 2\pi f_c t_1,\ M^*(t)\cos 2\pi f_c t_2\}$$

$$= \mathbb{E}\{M(t_1)M^*(t_2)\}\{\cos(2\pi f_c(t_1 - t_2)) + \cos(2\pi f_c(t_1 + t_2))\}$$

→ How do you get this in a Computer
→ How to Simulate

mean and Rx is periodic

↑

Q $\boxed{\text{Cyclostationary}}$ → the above this is periodic

- Sensors use this (CSMA) to see if a band is occupied or not

- $X(t)$ is periodic with $\frac{1}{f_c}$

- How to say it is periodic $R_{XX}(t_1+T_1, t_2) = R_{XX}(t_1, t_2)$
$= R_{XX}(t_1, t_2+T_2)$

$T_1 = T_2$ for cyclostationary

- What happens to the statistic of a random process after going through LTI system

$X(t) \to \boxed{LTI} \to Y(t)$ $\qquad Y(t) = h(t) \circledast X(t)$

$h(t)$

$\mathbb{E}\{Y(t)\} = \mathbb{E}\left\{\int X(\tau)\, h(t-\tau)\, d\tau\right\}$ ← looking at mean first

- we can swap ~~integrals~~ expectation if integrals are defined

$\Rightarrow \int \mathbb{E}(x(\tau))\, h(t-\tau)\, d\tau$

Not random

$t-\tau$ doesn't matter

$\mathbb{E}\{Y(t)\} = m_x(t) \int_{-\infty}^{\infty} h(t)\, d\tau$ → if system is unstable we cannot say anything

→ not same as $\int_{-\infty}^{\infty} |h(t)| < \infty$ → $H(0)$ ($f=0$)

if system is not stable. our expectation becomes ∞.

- defining cross correlation

$R_{XY}(t_1, t_2) = \mathbb{E}\{X(t_1)\, Y^*(t_2)\}$

- if X, Y are WSS and if $R_{XY}(t_1 - t_2)$ (τ)

they are jointly WSS

- We again have a property $R_{XY}(\tau) = R_{YX}^*(-\tau)$

↳ X and Y are even and real then $R_{XY} = R_{YX}$

New class

•) $R_{XY} = \mathbb{E}\{X(t+\tau)\,Y(t)\}$ → if X, Y are real

$R_{YX} = \mathbb{E}\{Y(t+\tau)\,X(t)\}$

•) when are the above things equal → When again?

•) how do you say two random ~~process~~ Variables are same

1 ↳ Strict equality → basically duplicate

2 ↳ Almost sure $X(w) = Y(w)$ when $P(w) \neq 0$

3 ↳ Same in distribution X and Y can ~~repeat~~ represent multiple things but their distribution is same.

→ we are talking about 2,3 in our case of R_{XY}

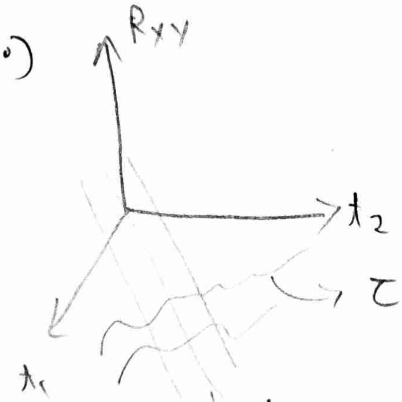
$Y(t+\tau)\,X(t) \Rightarrow Y(-t-\tau)\,X(-t)$

$\downarrow$ V $\quad$ $\downarrow$ $V+\tau$ $\Rightarrow Y(V)\,X(V+\tau)$

$\Downarrow$ equal to above first thing

•) X was τ ahead of Y, Same here after flipping

•)



// Check again

↳ τ line, we only see a 1-D curve instead of 3-D

↳ it is Same along here (these lines)

- $R_{xx}(\tau) = \mathbb{E}\{X(t+\tau)X(t)\}$ → for real

$R_{xx}(-\tau) = \mathbb{E}\{X(t-\tau)X(t)\}$

- $X(t) \to \boxed{LTI} \to Y(t)$
  $h(t)$

$$R_{YX}(t_1,t_2) = \mathbb{E}\left\{X^*(t_2)\overbrace{\int X(t_1-\tau)h(\tau)d\tau}^{Y(t_1)}\right\}$$

$$= \int \mathbb{E}\{X(t_1-\tau)X^*(t_2)\}h(\tau)\,d\tau$$

$\cancel{R_{xx}(\ )*h(\ )}$ $R_{YX} = R_{xx}(v) * h(v)$

$= R^*_{YX}(-\tau)$

- $R_{XY} = R^*_{xx}(-v) \circledast h(-v)$ → statistics of random process are also convolved with $h(t)$ → Because of linearity

$= R_{xx}(v) \circledast h(-v)$

- $R_{YY}(\tau) = R_{xx} * h(v) * h^*(-v)$

New class

- $X(t), Y(t)$ are WSS and $R_{XY}(t_1-t_2) = R_{XY}(\tau) = R^*_{YX}(-\tau)$ then X, Y are jointly WSS

- $\cancel{R_{XY}} = R_{YX} = R_{xx}(\tau) * h(\tau)$ → this ~~does~~ doesn't tell us $Y(t)$ is WSS

- $R_{XY} = R_{xx}(\tau) \circledast h^*(-\tau)$

- $R_{YY} = \{Y(t_1)Y^*(t_2)\}$

$$= \mathbb{E}\int X(t_1-\tau)h(\tau)d\tau\, Y^*(t_2)$$

$$= \int R_{XY}(t_1-\tau-t_2)h(\tau)\cdot d\tau \qquad \text{take } t_1-t_2 = V$$

$$= \int R_{XY}(V-\tau)h(\tau)\,d\tau$$

$$R_{YY} = R_{XY}(V) \circledast h(V) = R_{XX}(V) * h^*(-V) * h(V)$$

- $R_{YY} = R_{XX}(v) * h^*(-v) * h(v)$

- Looking at the spectrum of a random process

- Say $X \in \mathbb{R}^n \sim N(\mu, \underbrace{\Sigma}_{\text{Co-varience}})$ gaussian random vector

- $Y = \underbrace{F}_{\text{DFT Matrix}} X$

  DFT Matrix, Y is also gaussion $\sim N\left(F\mu, F\Sigma F^H\right)$

  If $\Sigma = \sigma^2 I \rightarrow$ i.i.d

  then the distribution of Y is also gaussion

  F is a unitary matrix (Just rotates)

- Instantanious spectrum is just random.
- $\mathbb{E}\{|X(f)|^2\} \rightarrow$ PSD (Power Spectral Density)

  $= \mathcal{F}\{R_{XX}(\tau)\}$ $\rightarrow$ Wiener–Khinchin Theorem

  $\hookrightarrow$ if it is WSS

- ~~Oscillators~~ Oscilloscopes take and show PSD $\rightarrow$ they cannot show phase of Fourier transform.

- When does the fourier transform gurantee real and positive output? as we are doing $\mathbb{E}\{|\cdot\cdot|^2\}$ $\rightarrow$ Conjugate Symmetric

- $|X(f)|^2 = X(f)\, X^*(f)$

  $= \int X_1(t_1)\, e^{-j2\pi f t_1}\, dt_1 \int X_2^*(t_2)\, e^{+j2\pi f t_2}$

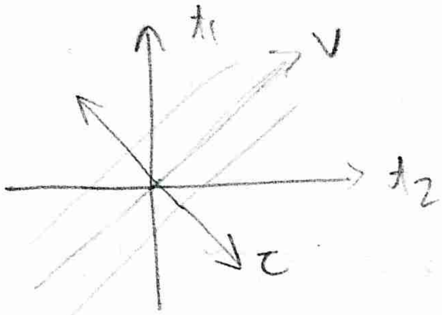
  $= \int x(t_1)\, x^*(t_2)\, e^{-j2\pi f(t_1 - t_2)}$

$E\{|X(f)|^2\} = \iint R_{XX}(\tau)\, e^{-2\pi f j}\, dt_1\, dt_2$

•) $x(t)$ $\quad x^*(-t)$ $\rightarrow$ $x(t) * x^*(-t)$ will always have a non negative transform

$\downarrow \qquad \downarrow$

$X(f) \qquad X^*(f)$



$t_1 - t_2 \qquad t_1 + t_2$

transform from t_1, t_2 to τ, V

$\tau \perp V$

we get

$$\begin{bmatrix} \tau \\ V \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}}_{J} \begin{bmatrix} t_1 \\ t_2 \end{bmatrix}$$

$$= \iint R_{xx}(\tau) e^{j2\pi f t} \frac{d\tau . dV}{2}$$ $\rightarrow$ ~~wed~~ what about limits?

$t_1 - t_2$, $\lim_{t_1, t_2 \to \infty}$ is what?

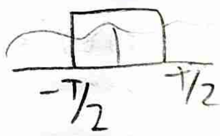
•) But here we get ∞ in the integration

•) So we normalize

$$\lim_{T \to \infty} \frac{1}{T} \mathbb{E}\{|X_T(f)|^2\}$$, where $X_T(f) = \mathcal{F}\{X(t) . Rect(\frac{t}{T})\}$

$\downarrow$

This is our definition of PSD



•) $N(t) \sim N(0, \sigma^2)$

$m_N(t) = 0$, $R_{NN}(t_1, t_2) = \delta(\tau)$ $t_1 \neq t_2$ $\mathbb{E} = 0$

Quiz 1

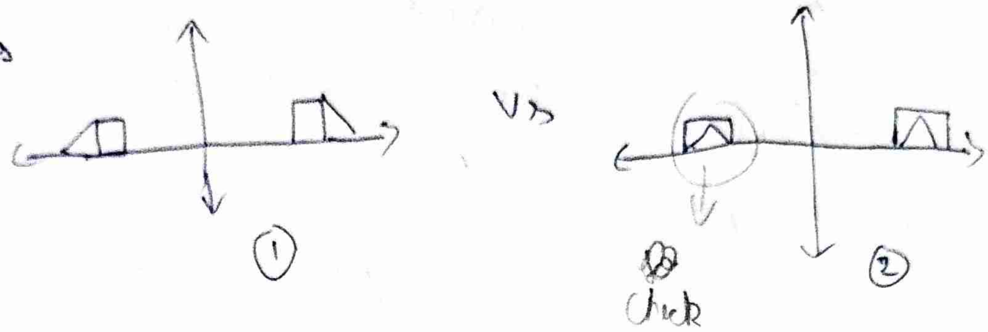
Q1 $\frac{1}{2\pi} \frac{d\phi(t)}{dt} = f(t) = \phi(t)$

Q2 $\Rightarrow$ Can have a ~~highloss~~ highpass filter at a higher freq

↳ Consider carrier freq as zero $\rightarrow$ modulate ~~of~~ FM signal with that assumption. hilbert transform

$x(t) = (0 + k_f \int_0^{\tau} m(\tau) d\tau)$ $\rightarrow$ then transport $x(t) + j x_h(t)$

Q3 Signals



Both have same bandwidth efficiency.
In real life we use ②, because hardward is simple. Only one oscillator

Q5 In signal power → if signals are orthogonal you can just add their ~~phase~~ power

Q6 $Y(t) = M(t)\cos(2\pi f_c t + \theta) + N(t)$

$\underbrace{\underset{WSS}{M(t)} \;\; \underset{\text{We know it is WSS}}{\cos} \;\; \underset{Unif}{\theta}}_{R_{XX}(\tau)} \qquad \underbrace{\text{i.i.d } N(0, \sigma^2)}_{\text{also WSS}}$

they are independent so you can add them

∴ ~~Y(t)...~~

∴ $Y(t)$ is WSS

• Ideal LPF, $h(t) * Y(t)\cos 2\pi f_c t$ is still WSS

• New class
Why is Autocorrelation a Power signal → It is a 2D signal only for a WSS signal ~~which~~ it is a 1D signal

• Auto correlation function can never be zero → implies energy is zero.

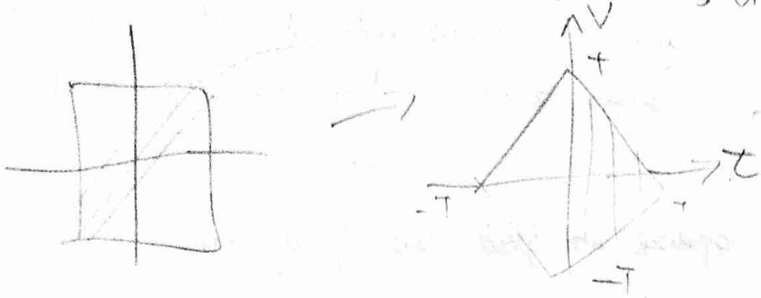
• Is there a signal whose $\int R_{xx}(\tau) = 0$ → $\cos(2\pi f_c \tau)$ → in the limit of ∞

• We look at auto correlation in a ~~window~~ window, for defining PSD

we have $\lim_{T\to\infty} \frac{1}{T} E\{|X_T(f)|^2\} = \int_{-T/2}^{T/2}\int_{-T/2}^{T/2} R_{xx}(t_1 - t_2)\, e^{-j2\pi f_c (t_1 - t_2)}\, dt_1\, dt_2$

$$= \lim_{T\to\infty} \frac{1}{T} \int_{-T}^{T} \int_{-T+|\tau|}^{T-|\tau|} R_{X_T X_T}(\tau)\, e^{-j2\pi f\tau} \frac{dv\, d\tau}{2}$$

→ Not WSS, only when $T\to\infty$



→ We have just rotated it

$\tau = t_1 - t_2$

$v = t_1 + t_2$

New class

- Wiener-Khinchin Theorem

PSD of $X(t) = \mathcal{F}\{R_{xx}(\tau)\}$

will it work if X is SSS → Yes (Because it is still WSS)

SSS says their $\left(X_1(t_1),\ \cancel{X_2(t_2)}\ X_1(t_1-d)\right)$ have the same distribution

we had then $\lim_{T\to\infty} E\left\{ \frac{1}{T} \int_{-T/2}^{T/2}\int_{-T/2}^{T/2} X_1(t_1) e^{-j2\pi f_c t_1}\, X_2^*(t_2) e^{j2\pi f_c t_2}\, dt_1\, dt_2 \right.$

→ limits come from $\text{rect}(t/T)$

but $X_1(t) = X(t)\ \text{rect}(t/T)$ → this is not WSS

i) $X_1(t)$ and R_{XX} need to satisfy dirichlet conditions → then we can take the expectation inside.

$$\Rightarrow \lim_{T\to\infty} \frac{1}{T}^{?} \int_{-T}^{T}\int_{-T+|\tau|}^{T-|\tau|} R_{xx}(\tau)\, e^{-j2\pi f\tau}\, \frac{dv\, d\tau}{2}$$

$$= \lim_{T\to\infty} \frac{1}{T}^{?} \int_{-T}^{T} R_{xx}(\tau)\, e^{-j2\pi f\tau}\, d\tau \times \left(\frac{2T-2|\tau|}{2T}\right) \to 1-\frac{|\tau|}{T} \to$$

↓ fourier transform

$\frac{1}{T}\text{rect}\left(\frac{\tau}{T/2}\right) \circledast \text{rect}\left(\frac{\tau}{T/2}\right)$

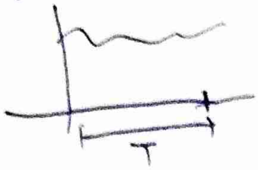
check

$\lim_{T\to\infty} R(f) * T\,\text{sinc}^2(fT)$

as $T\to\infty$, this tends to $\delta(f)$, we get $R(f) * \delta(f) \Rightarrow R(f)$

·) How to find PSD & IRL

$X(t)$:



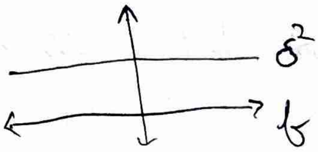
take $\int x(t)\, x^*(t-1)$ in different windows and correlate with it self with the same delay → then average over
This is the expectation

·) Small window may not capture the low freq signals in that window, Large windows are never a problem → you do not need to go beyond the lowest freq.

·) AWGN Model

$Y(t) = X(t) + N(t)$, $N \sim \mathcal{N}(0, \delta^2)$

$\therefore R_{NN}(\tau) = \delta^2 \delta(\tau) \rightarrow \mathcal{F} \rightarrow$



it is present in all frequencies → white

·) But this has ∞ Power → SNR=0

·) $SNR = \dfrac{\text{Power } x(t)}{\text{Power } N(t)}$

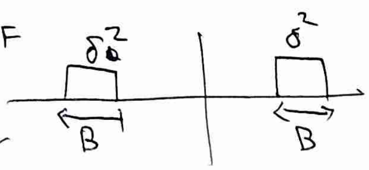
→ In real life there is always a filter in our hardware so we never observe the full spectrum of the Noise

$Y(t) = (X(t) + N(t)) * \overset{\rightarrow \text{filter}}{h(t)}$

·) We are starting with an ~~uncore~~ ~~uncorrel~~ uncorrelated function
↳ $\delta(\tau)$ but after you filter the Correlation changes.
The samples are still from gausian distribution but they are correlated

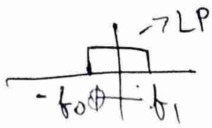
·) LTI System Cannot Convert an uncorrelated signal to correlated

Spectrum of noise after BPF



↘ Power → integrate with f

Power = $\delta 2B$



$B = f_0 + f_1$

Power = δB

New class

Why do we define ~~δ~~ $R_{NN}(\tau) = \delta(\tau)$ instead of $\begin{cases} 1 & \tau=0 \\ 0 & \tau\neq 0 \end{cases}$

$\delta(\tau) \downarrow$ Power is not zero

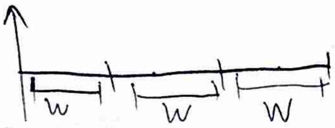
$\downarrow$ you get Power 0

$\downarrow$ Lim = 0 at $\tau = 0$

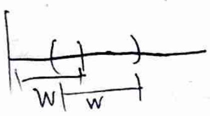
But we have 1 at $\tau=0$

We are looking at Band limited Power spectrum so we want it to be $\delta(\tau)$

- Computing PSD

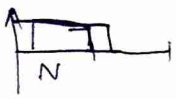


$\rightarrow$ Take FFT for W samples and do average (Bartlett)



$\rightarrow$ overlapped window (Welsch) then take FFT

~~In~~ IRL

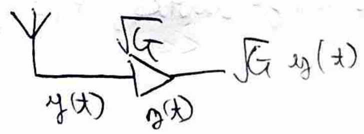


~~Rxx~~ $R_{xx}(m) = \frac{1}{N}\sum_{i=1}^{N} x[i]x[i-m]$ $\rightarrow$ after this do fft.

- Say $R_{NN}(\tau) = N_o\,\delta(\tau)$, $N_0 = KT \rightarrow$ Noise

Noise Power $= N_o B$ $\hookrightarrow$ Bandwidth (including -ve freq)

- ~~Does~~ An amplifier reduces SNR



$\sqrt{G}(y) = \sqrt{G}\,\underset{\downarrow\, P_s}{x(t)} + \sqrt{G}\,n(t)$

$z(t) = \sqrt{G}\,y(t) + N_1(t)$

$\rightarrow$ Noise added by amplifier

$SNR_{output} = \dfrac{GP_s}{GN_oB + N_1B}$ $\rightarrow$ Power adder $\rightarrow$ Independent.

$R_{yy} = E\left\{ (X(t_1) + N(t_1))(X^*(t_2) + N^*(t_2)) \right\}$

FFT $\downarrow$ $= R_{XX} + R_{NN} + R_{XN} + R_{NX}$

$S_y = S_X + S_N + C_{XN} + C_{NX}$ $\rightarrow$ Cross ~~pro~~ power spectral density

$\hookrightarrow$ Can be complex as well $(R_{XY}(\tau) = R^*_{YX}(-\tau))$

$C_{XN} + C_{NX}$ together is real

Conjugate Symmetric

Check

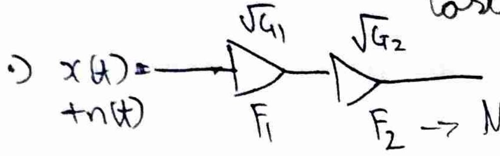
- ~~Noise figure~~ Noise factor $= \dfrac{SNR_{i/p}}{SNR_{o/p}} \geq 1$

- Noise figure (dB) $= NF = SNR_{i/p} - SNR_{o/p}$

  $10 \log_{10}(P)$ — write dBm, dBW

- Open WRT to change transmit power of any router

Cascade ↓



$$\sqrt{G_1}\,x(t) + n_1(t) + \sqrt{G_1}\,n(t) \to \sqrt{G_1 G_2}\,x(t) + n_2(t) + \sqrt{G_1 G_2}\,n(t) + \sqrt{G_2}\,n_1(t)$$

$$\underset{\text{output}}{SNR} = \frac{G_1 G_2 P_s}{G_1 G_2 N_0 B + G_2 N_1 B + N_2 B}$$

$$F = \frac{SNR_{i/p}}{SNR_{o/p}} = \frac{P_s / N_0 B}{\dfrac{G_1 G_2 P_s}{G_1 G_2 N_0 B + G_2 N_1 B + N_2 B}} = \frac{1}{N_0 G_1 G_2 B} \times (G_1 G_2 N_0 B + G_2 N_1 B + N_2 B)$$

$$= 1 + \frac{N_1}{N_0 G_1} + \frac{N_2}{N_0 G_1 G_2}$$

- for a ~~gen~~ single device, $F_1 = \dfrac{P_s / N_0 B}{\dfrac{G P_s}{G N_0 B + N_1 B}} = 1 + \dfrac{N_1}{G N_0}$

$F_2 = 1 + \dfrac{N_2}{G_2 N_0}$ → Considering N_0 is given as input

using this we get $\boxed{F_1 + \dfrac{F_2 - 1}{G_1}}$

$$F_{cascaded} = F_1 + \sum_{i=2}^{n} \frac{F_i - 1}{\prod_{j=1} G_j}$$ Friis formula

- Can we use some LTI block to maximize SNR

$x(t) \to \boxed{h(t)} \to$

↓

Matched filter

$$SNR = \frac{P_x}{P_n} = \frac{R_{xx}(0)}{N_0 B}$$ $\int_{-\infty}^{\infty} S_x(f)\,df = R_{xx}(0)$

LNA
LPF digital

- A Low noise Amplifier (LNA) in very narrow band
- In a digital setting we can Increase SNR
  - ↳ → LPF ~ → This happens when the spectrum of noise is present outside the spectrum of the signal → we can reduce that to improve SNR, But in the same band you cannot remove noise.

- New class
- → What filter can we use to improve SNR

$$y(t) = x(t) + n(t)$$

$h(t) * y(t)$ → improves SNR

$$h(t) * y(t) = \underbrace{h(t) * x(t)}_{z(t)} + h(t) * n(t)$$

$$P_N = \sigma^2 \int |H(f)|^2 df \quad \text{for } n \sim N(0, \sigma^2)$$

$$R_{zz^*}(\tau)\Big|_{\tau=0} \Rightarrow \int_{-\infty}^{\infty} S_x(f)\,df = R_{xx}(0)$$

↳ Power from PSD

$$R_{zz}(\tau) = \mathbb{E}\left\{\left(x(t+\tau) * h(t+\tau)\right)\left(x^*(t) * h(t)\right)\right\}\Big|_{\tau=0} \rightarrow \text{Substitute}$$

Assuming R_{zz} (depends on τ) then t can be anything (put $t=0$)

$$R_{zz} = \mathbb{E}\left\{\iint \underbrace{x(v)\,h(-v)}_{g(v)}\; \underbrace{x^*(s)\,h^*(-s)}_{g^*(s)}\, ds\, dv\right\}$$

↓ Inner product, it is maximized $g(v) = g^*(s)$ → Check

So we have $h(t) = x^*(-t)$ → we cannot construct this as our message signal is random (If we know the message whats the point of communication)

- $x(t) = m(t) \cdot p(t)$ — $m(t)$: Random message; $p(t)$: Carrier, Deterministic

assumption $m(t)$ is constant in the interval where $h(t)$ is matched to $p(t)$

$$= \iint E\{m(t)\}\, p(v) h(-v) p^*(s) h^*(-s)\, ds\, dv$$

$h(t) = p^*(-t)$

Also we have already used Matched filter before → demodulation we multiplied the carrier where $p^*(-t)$ is the ~~same~~ carrier only
$e^{-j2\pi f_c t}$

•) $N(t) = N_I(t) + jN_Q(t)$ → N_I, N_Q (independent, Uncorrelated and ~~real~~ jointly WSS for N to ~~ee~~ be WSS)

$$R_{nn} = \mathbb{E}\left\{ \left(N_I(t_1) + jN_Q(t_1)\right)\left(N_I(t_2) - jN_Q(t_2)\right) \right\}$$

$$= \mathbb{E}\left\{ (N_I(t_1) N_I(t_2) + N_Q(t_1) N_Q(t_2)\right)$$

$R_{nn}(\tau) = R_{N_I}(\tau) + R_{N_Q}(\tau)$

↓

$N_0 \delta(\tau)$, if the noise in IID, then $R_{N_I} = R_{N_Q} = \frac{N_0}{2} \delta(\tau)$

↓

$\sim \mathbb{C}N(0, N_0)$ ↳ complex gaussian

$\sim N(0, \frac{N_0}{2})$ → real Gaussian

•) ~~To~~ If LO is not synchronized then power in individual arm is different

•) SNR in the Passband

↳ after demodulation we have $A_c m(t) + n(t)$ $\quad SNR = \frac{A_c^2 P_m}{N_0 B}$ → Bandwidth

Noise Power in In Passband

$N_p = N_I(t) \cos 2\pi f_c t - N_Q(t) \sin 2\pi f_c t$ → this is always real

↓

Cyclostationary individualy but together they are not

~~If~~ is N_p WSS? $\quad E\{N_p\} = 0$

$$R_{N_p} = \mathbb{E}\{ N_p(t_1) N_p(t_2) \}$$

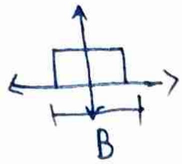
$$R_{N_P} = \Big\{ N_I(t_1) N_I(t_2) \left[\cos(2\pi f_c (t_1 - t_2)) + \cos 2\pi f_c (t_1 + t_2) \right] + N_Q(t_1) N_Q(t_2) \left[\cos(2\pi f_c (t_1 - t_2)) - \cos 2\pi f_c (t_1 + t_2) \right] \Big\} \to ⓐ$$

\# So $R_{N_P} = E\{ \frac{1}{2} ⓐ \}$, as noise are identical

$= \frac{1}{2} \left[R_{N_I} + R_{N_Q} \right]$ $\quad t_1 - t_2 = \tau$

New class

- we can put a LPF or BPF after we recieve to get rid of $t_1 + t_2$ terms
- Say $N_I = N_R(t) * h(t) \to$



$\to$ Power $= \frac{N_0}{2} B$

$N_R(\tau) = \frac{N_0}{2} \delta(\tau)$

$$R_{N_P} = (R_{N_R}(\tau) * h(\tau) * h^*(-\tau)) \times \cos 2\pi f_c \tau$$

$\hookrightarrow \frac{\delta(f - f_c) + \delta(f + f_c)}{2}$

$\to$ Power $= \frac{N_0}{4} \times 2B = \frac{N_0}{2} B$

The power remains the same? Its just that the components are not orthogonal so its not like their power adds up.

- AM Signals

$A_c m(t) \cos 2\pi f_c t + N_p(t)$

In the base band we have $A_c m(t) + N_I(t) \to$ in the $\cos 2\pi f_c t$ or I arm

$SNR_p = \frac{A_c^2 P_m / 2}{N_0 B / 2}$, $SNR_b = \frac{A_c^2 P_m}{N_0 B/2} \to$ DSB-SC

Convolution with cos cos add the power up.

DSB-AM

$A_c(1 + \alpha m(t)) \cos 2\pi f_c t + N_p(t)$

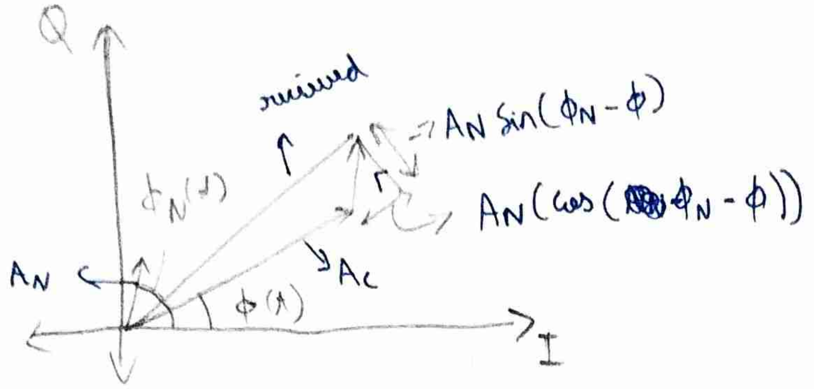
$SNR_p = \frac{\frac{A_c^2}{2}(1 + \alpha^2 P_m)}{\frac{N_0 B}{2}}$ $\qquad SNR_b = \frac{A_c^2 \alpha^2 P_m}{N_0 B / 2}$

$= \frac{2\alpha^2 P_m}{1 + \alpha^2 P_m} SNR_p$

Baseband SNR drops a lot for $\alpha < 1$, this is because a lot of Power is used in the carrier.

FM

$A_c \cos(2\pi f_c t + \phi(t)) + N_p(t)$
$\hookrightarrow$ convert to $A_N(t)\cos(2\pi f_c t + \phi_n(t))$



$$y(t) = \sqrt{(A_c + A_N\cos(\cdot\cdot))^2 + A_N^2\sin(\cdot\cdot)}\ \cos\left(2\pi f_c t + \phi(t) + \tan^{-1}\left(\frac{A_N \sin(\cdot\cdot)}{(A_c + A_N)\cos(\cdot\cdot)}\right)\right)$$

~~if we assume~~ Our message info is present only in the phase.

If $A_c >> A_N$

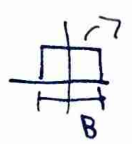
$$\approx \phi(t) + \frac{A_n(t)}{A_c}\sin(\phi_n(t) - \phi(t)) \rightarrow \text{SNR increases with } A_c$$

~~assume~~

$$\Rightarrow \frac{1}{A_c}\left\{\left[A_N \sin\phi_n(t)\cos(\phi) - \cos(\phi_n(t))\sin(\phi)\right]\right\}$$

assume $\phi_n(t)$ varies a lot more that $\phi \rightarrow$ so assume constant

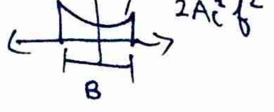
this is ~~called~~ just rotation so



$\Rightarrow$ for FM we take a derivative of phase

$|H(f)|^2 = f^2$

$\frac{1}{2\pi}\frac{d}{dt}$

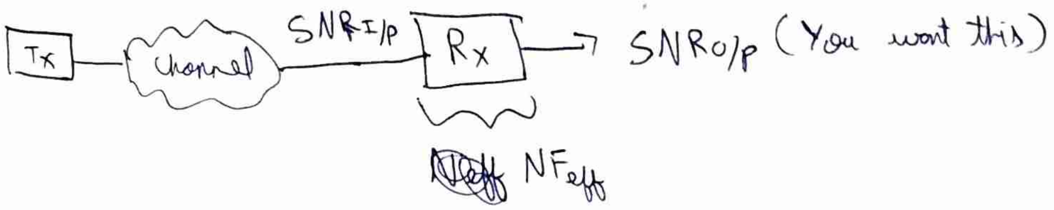


$$\frac{N_0}{2A_c^2}\int_{-B/2}^{B/2} f^2\, df = \frac{B^3}{12}$$

$SNR_{FM} = K$ Photo on Phone

New ~~class~~ class

Link budget analysis



- $y(t) = h\,x(t) + n(t)$
  - $h \downarrow$ ~~att~~ attenuation by channel

- Propagation loss (for free space) $= \left(\frac{4\pi d}{\lambda}\right)^2 = \frac{P_{Tx}}{P_{Rx}} \Rightarrow P_{Rx} = \frac{P_T}{\left(\frac{4\pi d}{\lambda}\right)^2}$ → inverse square law

- We can find NF_{eff} for Channel + Rx

$SNR_{o/p} = 5dB$ → $SNR_{ip} = NF + SNR_{op}$

$T_{xp} = 60\,dB + P_s$ → 15 dBm

60 dB ↓ 1 Km away, 4π wavelength.

Digital Communication

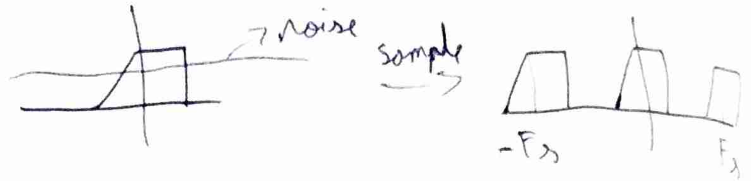
→ allows us to communicate at -ve SNR as you can map back to bits.

$x(t) \longrightarrow x[n]$

$\downarrow$

B $\qquad F_s > 2B$ (Nyquist)

•) What if there is noise?



does the noise power go to ∞?

•) No because you have to calculate PSD

•) In Practice we add an anti aliasing filter

$x[n] + N[n]$

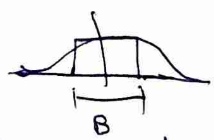
$R_x[\tau] \rightarrow \delta[\tau] \xrightarrow{DFT} PSD$

$\downarrow$ $\qquad\qquad\qquad\qquad \downarrow$

B $\qquad\qquad\qquad$ or DTFT Check?

•) Noise effective Bandwidth

after a filter ↳



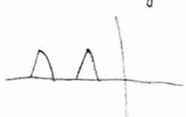
there is noise outside B which all aliases when we sample

We just compute

$$B_N = \frac{\int_{-\infty}^{\infty} S_N(f)\,df}{S_{max}(f)} \qquad N_0 = KTB_N$$

•) What should we sample this at

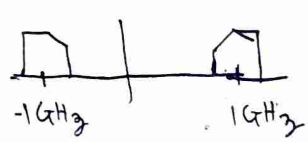
↳ there is a range



↓

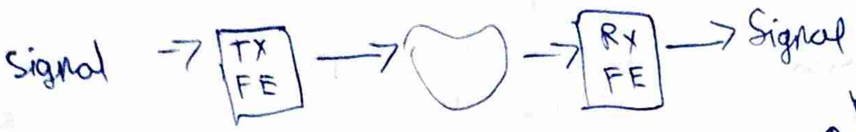
Band Pass Sampling

P

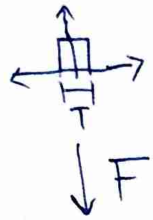


New class

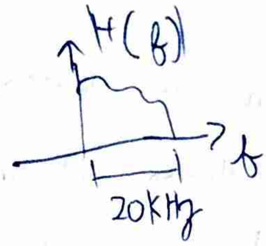
- About the competition



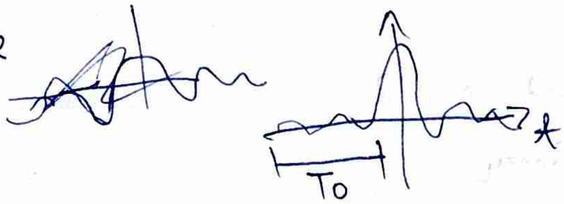
~~BPSK~~ BPSK



we want to ~~up~~ use



we want to pick



Sink ~~spectrum~~ with

- ~~90~~ 99% energy, we have a latency of To as we need to know 0 or 1 before transmitting (Non Causal).

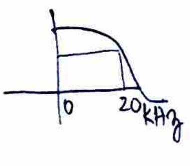
- We estimate the $\hat{H}(f)$ at the reciever, find $\hat{h}^{-1}(t) = f^{-1}(\hat{H}^{-1}(f))$ then ~~w~~ you are sending

$$x(t) = \sum_{k=0}^{\infty} b[k] * \underbrace{p(t-nT)}_{\text{Pulse}} \quad , \text{ we send } \underbrace{x(t) * \hat{h}^{-1}(t)}_{\text{Precoding}}$$

why at the ~~reciever~~ transmitter? because we convolve with noise at the reciever

→ do at transmitter because reciever will have noise

- You can do matched filtering with a pulse shape
- ~~10~~ to find $\hat{H}^{-1}(f)$ you sent ~~a~~ a $\delta(t)$ but we canot send this so we will send sinc(t) where ~~pulse~~ is in frequency



we get something above 20KHz

- But we have currently only ~~20Kb~~ 20 kbps, to increase this, we can ~~g~~ use ~~the~~ complex signals. But in real domain only we can have more levels of pulse. 4 levels 00, 01, 10, 11
- we can only 16 levels or 8 bits max → quantizer of reciever

·) ~~Can you~~ ^can^ ML $x(t) \rightarrow \boxed{Tx - Chn - Rx} \rightarrow y(t)$

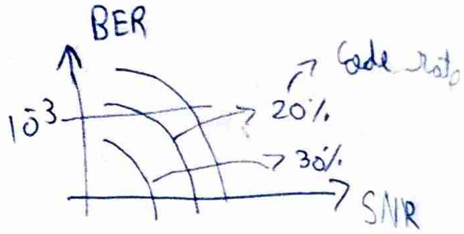
you can use ML

train with regression $y(t) \rightarrow \boxed{} \rightarrow x(t)$

·) Let us try for BER(Bit error rate) $\leq 10^{-3}$

·) Try reed Solomon or hamming codes

·) How much Error coding do you need



pick a BER and then Based on SNR
pick your ~~reduced~~ redundancy

·) Synchronize the transmitter and receiver → Send a pseudo random sequence where auto correlation is zero ~~on~~ if shift is even a small amount.

Quiz 2

Q1 a) KTB = $\underline{-101 \text{ dBm}}$

b) SNR = $\underline{101 \text{ dB}}$

c) SNR o/p = 91dB

d) $L = \left(\frac{4\pi d}{\lambda}\right)^2 \approx 60\text{dB}$

$$24 - 60 = -36 + 10\log \underset{\text{Noise}}{N}$$

Noise = thermal + interference
thermal → negligible

e) $F_1 + \frac{F_2 - 1}{G_1} \approx F_1$

Q2 Cyclostationary process $R_x(t+T, \tau) = R_x(t, \tau)$
↳ it is two dimentional

average it over τ to get a periodic (Cyclostationary) process
↳ ~~R(t)~~ $\bar{R}(t)$, $\bar{R}(\tau)$ is average over ~~T~~ t
↳ it is not WSS

↳ have to check the mean is also periodic

Q3 ~~doesn~~ doesn't matter if we measure noise in Passband / baseband

$N(t) = N_I \cos 2\pi f_c t - N_Q \sin(2\pi f_c t) = A(t)\cos(2\pi f_c t + \theta(t))$

$N_I + j N_Q$

$\to$ Variance of $\theta \sim \text{Unif}[-\pi, \pi]$

A $E[\theta(t+\tau)\,\theta(t)] = \frac{\pi^2}{3}\delta(\tau)$

$\to$ mean of A^2

$E\{(N_I^2(t+\tau) + N_Q^2(t+\tau))(N_I^2(t) + N_Q^2(t))\}$ $\to$ as they are IId

$= 2E\{N_I^2(t+\tau)N_I^2(t)\} + 2E\{N_I^2(t+\tau)N_Q^2(t)\}$

$\downarrow$ discussed below - (a)

$=$ ~~$2\sigma^4$~~ 2 (curtosis something)

$\to$ Band Unlimited

$= 2 \left(\begin{matrix} 3\sigma^4 & \tau = 0 \\ \sigma^4 & \end{matrix} \right.$ $\to$ 4 th order momentum

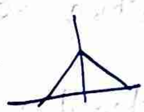
$= 4\sigma^4(\delta(\tau) + 1) \to$ PSD is the F.T of this

$\hookrightarrow$ they are not a zero mean distribution so it is not a $\delta(t)$,

N_I^2 is zero mean but $N_I^2 + N_Q^2$ is not

$\cdot$) The above is for bandunlimited N_I, N_Q,

Band limited $\downarrow$

$N_I \to$  $\to N_I^2 =$ 

$N_I^2 + N_Q^2 \to$ the PSD doesn't add UP, the fourier transform of the signal does but not PSD

$\hookrightarrow$ this doesn't go to zero even if these signals are independent, because they might not have a zero mean $\to$ cross terms are not zero

think of this ~~as~~ in terms of Jensens inequality $f(x(t)) = x^2(t)$

$\therefore E[f(N_I), f(N_Q)] \to$

-(a) $\to 2E\{N_I^2(t+\tau)\}E\{N_Q^2(t+\tau)\} = 2R_{N_I}^2(0)$

Now how to handle $E\{N_I^2(t+\tau)N_I^2(t)\}$

Isserlis theorem

$$E\{X_1X_2X_3X_4\} = E[X_1X_2]E[X_1X_3] + E[X_1X_2]E[X_2,X_4] + E[X_1X_4]E[X_2X_3]$$

after solving $4N_0^2(R^2(\tau)+R_N^2(0))$

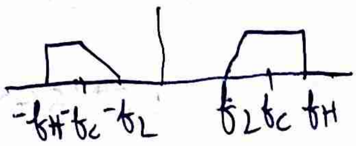
↳ this is the Δ + additional stuff

Q 4 $X(t)$ we get it at $t=nT_s$

$$\underbrace{S[k]}_{1} + \underbrace{N[k]}_{N_0}$$

$$-10\log N_0 \ge 4 \Rightarrow N_0 \le 3.98$$

Bandpass Sampling



we do not want overlap so we have these conditions

$-f_L + k f_s \le f_L$ the kth component shouldn't overlap

→ anything before this comes before

$-f_H + (k+1) f_s \ge f_H$ → anything after this comes after

if this is true for one component then it is true for all components

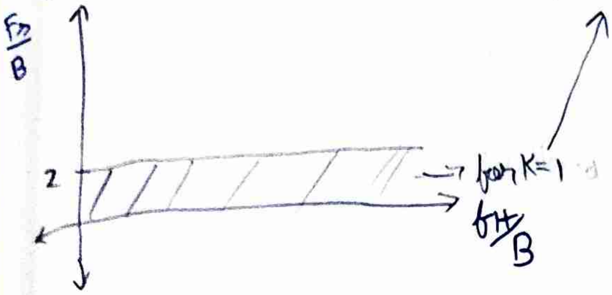
$$\Rightarrow \frac{2f_H}{K+1} \le f_s \le \frac{2f_L}{K}$$

$$\Rightarrow \frac{2f_H}{k} \le f_s \le \frac{2f_L}{K-1}$$ → need to find k

↳ $\frac{2f_H}{k} \le \frac{2f_L}{k-1} \Rightarrow \frac{1}{f_L} \ge K\left[\frac{1}{f_L} - \frac{1}{f_H}\right]$

or $k \le \frac{f_H}{f_H - f_L} \Rightarrow \boxed{k \le \frac{f_H}{B}}$

from the inequality $\quad \frac{2f_H}{kB} \le \frac{F_s}{B} \le \frac{2f_L}{(k-1)B}$



$$2 \le \frac{2f_H}{kB} \le \frac{F_s}{B} \le \frac{2f_L}{(k-1)B} = \frac{2(f_H/B - 1)}{k-1}$$

if $k=0$, when f_H is comparable to B → overlap occurs

if $k=1$, $\frac{F_s}{B} \le \infty$

now for every k, $2\left(\frac{f_H}{B}-1\right)$ is the upper bound

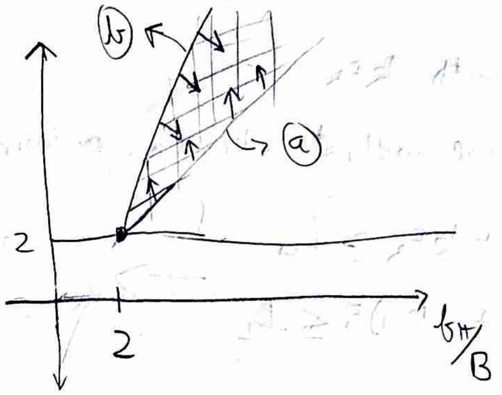
$2\left(\frac{f_H}{B}\right)$ is the lower bound.

if $f_H >> B \Rightarrow f_H \approx f_L$ or the ~~gap~~ gap between allowed regions for sampling reduces, small shift in LO causes aliasing

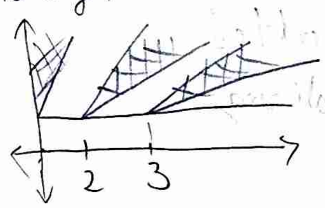
for $k=2$ $\quad \frac{2f_H}{2B} \le \frac{F_s}{B} \le \frac{2f_L}{B}$

(a) $\frac{2f_H}{2B}$

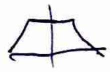
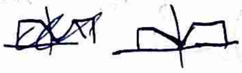
(b) $\frac{2f_L}{B} = 2\left(\frac{f_H}{B}-1\right)$



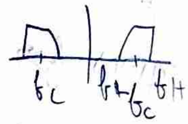
1) For any K



ask lekha

if we get  instead of  when Sampling how do we flip it ?

Newclass

- we need 2 parameters f_c and Bandwidth (B) or f_L and f_H 

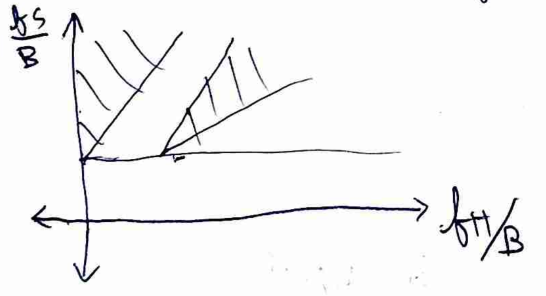
- Sample with kF_s

- $F_s \geq B$ we ~~with~~ is the bare minimum

$$-f_H + kF_s \geq f_H$$
$$-f_L + (k-1)F_s \leq f_L$$
$$\Longrightarrow \quad \frac{2(f_c + B/2)}{k} \leq f_s \leq \frac{2(f_c - B/2)}{k-1}$$

and $f_s \geq B$

- We ~~can~~ make a plot of 2 variables by normalizing

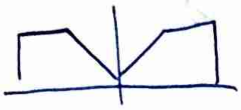


- How to find k
$$\frac{2f_H}{K} \leq \frac{2(f_H - B)}{k-1} \quad f_L$$

$$\Rightarrow k \leq \frac{f_H}{B}$$

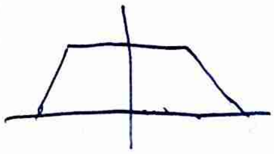
as at 2 4B ↳ ask manoj why

When you sample you get one of the two below, how do you convert one to another?

1)  $\to -f_L + nf_s = 0 \qquad nf_s = f_L \leq \frac{2nf_L}{k-1}$

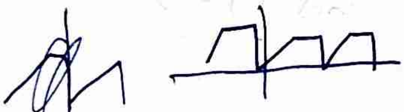
$k-1 \leq 2n$

$\to$ k odd $\to$ how check

2)  $-f_H + nf_s = 0 \qquad \to k$ even

$f_H = nf_s \geq \frac{2nf_H}{k} \to k \geq 2n$

•) Say you have

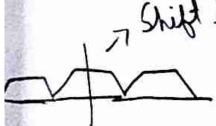
 how to get  $\to$ shift by $B/2$

$X(f) \qquad \downarrow X(f - B/2)$

Similarly

in ~~the~~ time domain multiply $e^{j2\pi t f_0} = e^{j2\pi n T_s B} \to f_0 = B$

$\to$ Shift by B to get

  $= (e^{j\pi})^n$ assuming $F_s = 2B$, $T_s = \frac{1}{F_s}$

↳ Check

$\downarrow$

+1 or −1 in time domain

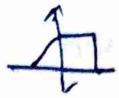
•) for  $\to e^{j2\pi \frac{n}{F_s} \times \frac{F_s}{4}} = (e^{j\frac{\pi}{2}})^n$

$\downarrow$

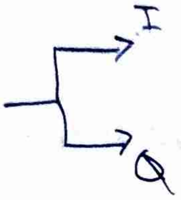
$1, j, -1, -j \ldots$

•) here you do not need to multiply by $\cos(2\pi f_c t)$ as we can just ~~So~~ multiply by complex exponential.

•) For noise $\to$ We ~~fst~~ first pass through BPF

•) if you want  but you receive 

this is because of I, Q arm getting different, the delay is different ?



→ I/Q ~~...~~ imbalance, $X_I + jcX_Q$
↓
Imaginary axis is flipped

•) After we do sampling there is Quantization

$\hat{x}[n] = Q\{x[n]\}$ → non-linear

•) Quantization causes loss of information

$\hat{x} = x + w$ → this is the quantization noise

$w = -\hat{x} + x$

$SQNR = \frac{P_x}{P_w}$ Can this be ∞ ? means $P_w = 0$

•) What if our input signal itself is discrete ? then w can be zero ? hmm

•) We want to ~~...~~ analytically evaluate SQNR

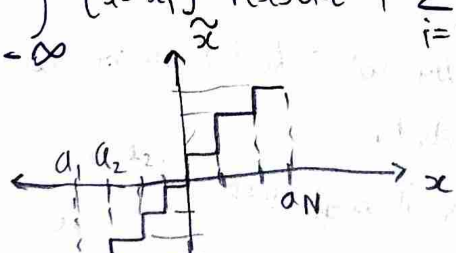
•) Assumptions → zero mean and uncorrelated

if it was correlated you will have to find PSD

•) $\mathbb{E}\{\underbrace{(x-\hat{x})^2}_{\text{Distortion}}\}$

•) $D = \int_{-\infty}^{a_1}(x-\hat{x}_1)^2 P(x)\,dx + \sum_{i=1}^{N-1}\int_{a_i}^{a_{i+1}}(x-\hat{x}_{i+1})^2 P(x)\,dx + \int_{a_{N-1}}^{\infty}(x-\hat{x}_N)P(x)\,dx$

distortion measure



- $N = 2^b$

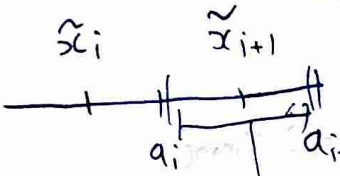
we can change $a_1 \dots a_{N-1}, \tilde{x}_1 \dots \tilde{x}_N$ to optimize our quantization ↳ it is discrete in time, we make it discrete in amplitude also

- $\tilde{x}_i$ can be 0000 or 1111 or how do you pick these

$$\frac{\partial D}{\partial a_i} = -(a_i - \tilde{x}_{i+1})^2 P(a_i) + (a_i - \tilde{x}_i)^2 P(a_i) = 0$$

$$\Rightarrow \boxed{a_i = \frac{\tilde{x}_{i+1} + \tilde{x}_i}{2}} \quad \text{— (a)}$$

→ independent of probability distribution



↳ the bounds are equidistant

- now $$\frac{\partial D}{\partial \tilde{x}_i} = -\int_{a_{i-1}}^{a_i} 2(x - \tilde{x}_i) P(x) = 0$$

$$= \boxed{\tilde{x}_i = \frac{\int_{a_{i-1}}^{a_i} x\, P(x)\, dx}{\int_{a_{i-1}}^{a_i} P(x)\, dx}} \quad \text{— (b)}$$

→ Conditional expectation

- ~~Day~~ Lloyd-Max optimizer → you need to alternate between (a) and (b) a_i and $\tilde{x}_i$ and keep optimizing untill you converge

this is scalar optimization

- You can also do ~~vector~~ optimization for higher dimensions, set of all ~~dis~~ points closest to any point

→ voronoi for x_3 → Used in Computer graphics

↳ needed to quantize complex numbers

→ ~~let~~ There is Uniform and non uniform quantization

let us fix $a_{i+1} - a_i = \Delta$

this implies → x_i is fixed (it is equidistant from a_{i+1} and a_i)

$$D = \int_{-\infty}^{a_1} (x - a_1 - \Delta/2)^2 P(x)\,dx + \sum_{i=1}^{N-1} \int_{a_1+(i-1)\Delta}^{a_1+i\Delta} (x - (a_1 + i\Delta - \Delta/2))^2 P(x)\,dx$$

$$+ \int_{a_{N-1}}^{\infty} (x - (a_{N-1} + \Delta/2))^2 P(x)\,dx$$

→ Let us fix $x \in [-x_{max}, x_{max}]$ → Dynamic range $= 2x_{max}$

$a_0 = -x_{max}$, $a_N = +x_{max}$

now $\Delta = \dfrac{2x_{max}}{N}$

if we assume $P(x)$ is Unif , $P(x) = \dfrac{1}{2x_{max}} = \dfrac{1}{\Delta N}$

Substituting this we get $D = N \displaystyle\int_{-\Delta/2}^{\Delta/2} y^2 \times \frac{1}{\Delta N}\,dy = \frac{\Delta^2}{12}$

How did we get y^2

New ~~class~~ class

now $SQNR = \dfrac{\Delta^2}{12} = \dfrac{3N^2}{\left(\dfrac{x^2_{max}}{P_x}\right)} \approx \dfrac{3 \cdot 4^b}{\alpha}$ → PARP

SQNR in dB $= 4.8 + 6b - 10\log\alpha$ α is always > 1

→ 1 bit increase gives us 6 dB extra SQNR

now if $b = 0$ that means we have a constant as output, this is because we are still calculating error $(c - x_{max})^2$

- Our ~~ou~~ audio signals are gaussian distribution → our assumption that x is uniform is wrong.
- we need to transform our signal to look like uniform to keep our ADC ~~is~~ optimal

$x \longrightarrow f(x)$ → ~~Us~~ You need to transform

$x \downarrow N(\cdot)$; $f(x) \hookrightarrow u$

$\boxed{\text{A-law}}$, Box-Muller Transform

$\boxed{\text{Compânding}}$ (pand)

$$f(x) = \text{Sign}(x) \times \frac{1+\log(A|x|)}{1+\log(A)} \rightarrow \text{if } |x| \geq \frac{1}{A}$$

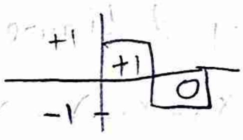
$$\text{Sign}(x) \times \frac{A|x|}{1+\log A} \rightarrow \text{else}$$

→ you multiply by Something → low numbers are higher vice versa

overall gaussian $\times$ $f(x)$ looks approx uniform

PCM (Pulse Code Modulation)

$x(t) \rightarrow x[n] \rightarrow Q(x[n]) \rightarrow \{b_i\}$



- In old CD, DVD ~~is~~ 1 hour of Audio ~~is~~ used to ~~be~~ be 600 MB

$40 \times 10^3 \times 3600 \times 8$ bit = 144 MB

$F_s = 40KHz$; 1 Hour

→ Check how? How to store

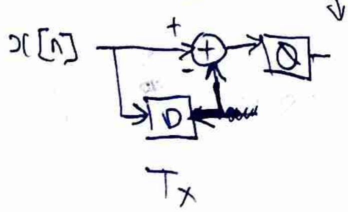
- MP3 file used to store audio in freq domain instead of time
- ML can be used to find non linear transform to compress other that F.T

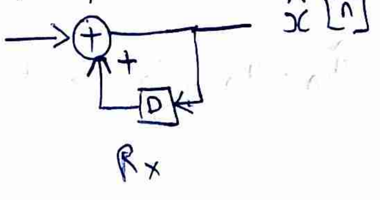
- How to change ↗ reduce Dynamic range (Change α to ↑SQNR)

Differential PCM → We made an assumption that the samples are uncorrelated → Wrong

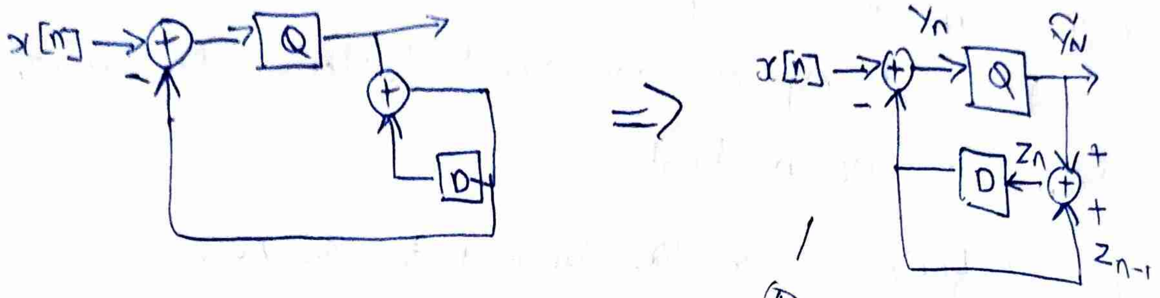
↓ this helps in ~~that~~ that

$x[n] - x[n-1]$





- We get a better performance in, we are doing $x_n - \tilde{x}_{n-1}$



ⓐ

new class

Idea for DPCM → $\max[x_n - x_{n-1}] < \max x_n$ — ⓑ

will this increase PARP?

Say $E\{x_n^2\} = P_x$ $\quad E\{(x_n - x_{n-1})^2\} = 2P_x - 2E\{x_n x_{n-1}\}$

error

$= 2P_x(1-\rho)$

we want ρ to be big, so ~~power~~ → error ↓ samples are close together

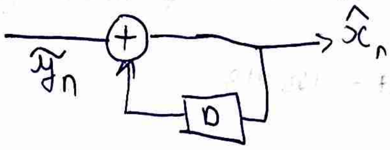
- Compressed sensing → to samples at lower than nyquist rate

- looking at figure ⓐ

$Q(x_n - x_{n-1}) = x_n - x_{n-1} + w_n$ → this keeps accumulating so we want to reduce this

we do $Q(x_n - \tilde{x}_{n-1})$ → it is reduced

- at the reciever we have



- Compare $\tilde{y}_n - y_n$ and $\hat{x}_n - x_n$

$\tilde{y}_n - y_n = \tilde{y}_n - x_n + Z_{n-1}$

$= Z_n - x_n$

$= \hat{x}_n - x_n$ $\quad \Rightarrow Z_n = \hat{x}_n$ as we have just copy pasted the reciever in ⓐ

This is true if $Z_0 = X_0 = 0$

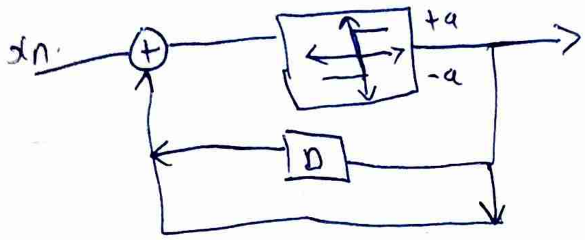
$\therefore \tilde{y}_n - y_n = \hat{x}_n - x_n$

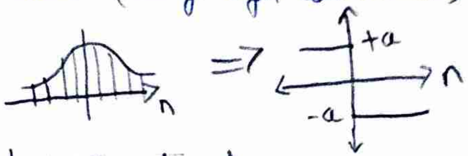
- But now SQNR of $(\tilde{y}_n - y_n)$ is > than SQNR $(\hat{x}_n - x_n)$

as from ⓑ

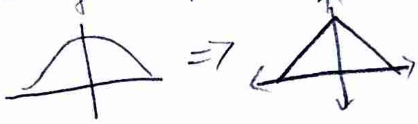
Delta Modulator

- we can do 1 bit ~~two~~ quantization by oversampling → ~~our~~ our signal looks like a straight line approx (very high correlated)





at high sample rate



- There is an slope overload, if signal has slope greater than $\frac{a}{T_s}$ then problem, so increase T_s

sample at T_s, max height $= a \times \frac{T}{T_s}$

Max slope $= \frac{a \times \frac{T}{T_s}}{T} = \frac{a}{T_s}$

$P(w_n = 0) = 0$

- $Y_n = X_n + W_n$, $\underbrace{\hat{X}_n = g(Y_n)}$ → $Pr(\hat{X}_n = X_n) = 0$ in analog com

estimate of X_n

- what if $x_n \in \{-1, 1\}$, and $g(Y_n) = \text{Sign}(y)$

$$Pr(\hat{X}_n = X_n) = \underbrace{Pr(X_n = +1)}_{\alpha} Pr(W_n > -1) + \underbrace{Pr(X_n = -1)}_{1-\alpha} \underbrace{Pr(W_n < 1)}_{\text{symmetric}}$$

$$= Pr(W_n > -1)$$

↳ greater than zero

- Signal Space analysis

$S_1(t) \dots S_N(t)$

⇓

$\psi_1(t) \dots \psi_N(t)$ (orthogonal signals)

then $\underbrace{\sum a_i \psi_i(t)}_{x(t)}$

a_i → message information

$\psi_i(t)$ → Pulse or carrier